The LLM-Attacker: A Unified Red-Team Framework for Privacy Evaluation of Split and Distributed LLM Systems

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Abstract

Split language models keep input text on a client but send intermediate representations to a server. Attacks can reconstruct text from these messages. Comparing privacy evaluations is difficult when attackers see different messages, know different model weights or use different datasets and recovery measures. We present LLM-Attacker, a framework that records these choices and lets researchers check, compare and extend an evaluation. Compatible attacks reuse recorded messages, and their predictions are saved for independent scoring. The scoring rule requires a linked undefended reference before a defence can pass. Reports also identify attacks that remain untested. In a controlled GPT-2 experiment, adding gradients returned during training increased token recovery from 94.20% to 97.38% and exact-document recovery from 13.71% to 37.77%. The 95% interval for the token gain lay above the two-percentage-point threshold fixed before testing. This experiment is shared with a companion study; here it tests how the evaluation responds to a change in attacker access. Further studies measure how corpus choice and compression defences affect recovery, compare an adapted published attack with its reference implementation, and test the measurement code. The framework reports implementation checks separately from measured attack effectiveness. Each privacy claim should state the attacker, dataset and scoring rule, report its controls and identify untested attacks.

1 Introduction

Split learning divides a language model between a client and a server [18]. The client keeps the input text and runs the first layers. It sends the server an *activation*, the intermediate representation computed at each token position. During training, the server returns a *gradient*, which describes how changes to these representations would affect the training loss. Both messages contain information about the private text.

Prior attacks reconstruct text from gradients [21, 3, 1], activations [11, 13] or both [2]. Their results depend on which messages the attacker sees, which model weights it knows and how much public text and computation it uses. The dataset and recovery measure also matter. A defence can appear effective when the attack already fails without it. An average token score can conceal exact recovery of complete documents or sensitive spans. Researchers need these details to interpret a privacy result and compare it with another study.

What must an evaluation record and test so that another researcher can check its conclusion, compare it fairly and extend it? LLM-Attacker records what the attacker can observe, what it knows and the resources it uses. Compatible attacks run on the same recorded messages. The framework requires a linked undefended reference before a candidate defence can pass its scoring rule, and reports missing tests explicitly. Saved predictions let other researchers check the scores without rerunning the target model.

The experiments test why these choices matter. Matched reconstruction comparisons measure the effect of adding gradients to the attacker's observations. Separate studies vary the dataset and defence. Comparisons with a published attack implementation check agreement on specified

inputs, while tests of the measurement code check scoring, controls and handling of the saved evidence. These checks establish different properties and are reported separately.

The contribution has three parts:

- **A common evaluation procedure.** We specify how to record the model, attacker access, data and resources, and how compatible attacks reuse the same observations.
- **Checkable privacy results.** We combine dataset-specific baselines, linked undefended controls and reports of missing tests, and retain predictions and configurations for independent scoring.
- **Evidence for the procedure.** We assess controlled reconstruction, agreement with a published implementation and the measurement code. The registry records the available validation evidence for all 64 modules, including three orchestration helpers.

This methodological contribution builds on two related studies. Politis and Pappas [15] audit a real two-node split-training system whose forward-message privacy check missed leakage through the returned gradient. Their study diagnoses that system's failure and tests observation coverage and probe calibration. A companion study by the same authors [16] examines how attacker observations and model knowledge affect per-token and per-document leakage. The controlled GPT-2 experiment reported here is shared with that study and provides one evidence base. Here we use it to assess the evaluation procedure alongside additional corpus, training-duration, defence and implementation studies. The broader Qwen and Mistral experiments provide further system and implementation checks under their own stated conditions.

2 An Explicit Evaluation Protocol

2.1 Threat model

The principal attacker passively observes the messages crossing one split. It sees activations during training and inference, and returned gradients during training. It neither modifies the traffic nor receives the private text, next-token labels or server layers. The capture harness computes the loss using the labels but does not expose them to the predictor. The scorer uses the text afterwards. This observation contract differs from an active server that controls returned tensors, and from a server that directly holds the loss labels.

By default, the attacker holds a *public copy* of the client's layers, the pretrained layers available for download. A stronger condition supplies the *trained copy*, including the client's fine-tuned embeddings. The attacker also receives public calibration text. Labelled-example, key-solving, active-server and side-channel modules declare their additional privileges separately. Supplementary Section S1 maps the observation surfaces and their associated modules.

2.2 A worked comparison

Consider the experiment at a split after GPT-2's sixth layer, with the attacker holding public client layers. We want to measure what returned gradients add to reconstruction. The model checkpoint, held-out documents and public calibration text used to fit the attacker stay fixed.

1. **Record the observations.** Run the target on the evaluation documents and save its activations and returned gradients. Give the activation-only attacker the activations. Give the joint attacker both messages. Neither attacker receives the evaluation text or next-token labels.
2. **Match the reconstruction search.** Both attacks consider 32 candidate tokens per position and keep two partial reconstructions at each search step. The joint attacker uses

gradients to propose some candidates. Shuffled-gradient and random-candidate controls test whether any gain depends on the gradient information. Matching the search does not match total computation; the joint attacker also ranks candidates using gradients.

3. **Score the saved predictions.** A separate scorer compares predictions with the held-out text. Token recovery rises from 94.20% to 97.38%; exact-document recovery rises from 13.71% to 37.77%. Exact-document recovery requires every token in a document to be correct. Save the predictions and settings so another researcher can check both measures.

This comparison isolates a change in attacker observations while fixing the stated search limits. It also shows why the recovery measure must be reported: the gain in complete documents is much larger than the token gain suggests. The statistical analysis and detailed attack mechanics appear in Sections 4.1 and 5.1.

2.3 Protocol, capture and replay

We describe an evaluation by

$$\mathcal{E} = (T, V, O, K, B, C, M, G). \tag{1}$$

Here T names the target model, split, defence and checkpoint; V the attacker's position and privileges; O its observations; K its knowledge; B the calibration, optimisation and search budgets; C the corpus, partitions and tokenisation; M the measured outcome; and G the controls, uncertainty calculation and decision rule. These components are recorded across the pinned protocol, run configuration and result records.

This definition identifies the experimental contrast. Reading gradients changes O, receiving trained layers changes K, and replacing Wikipedia with code changes C. An attack comparison holds the remaining components fixed and reports each attack's assigned budget. The joint-versus-activation experiment additionally matches candidate count and beam width, so its controls can isolate the gradient's information.

Capture separates target execution from attack execution (Figure 1). A split capture stores forward messages and, for training, returned gradients, together with their session and ordering metadata. Compatible attacks replay these observations against the same checkpoint and documents. Other surfaces, such as timing or model outputs, use their own observation providers. This arrangement supports paired comparisons, later attack additions and independent rescoring without retraining the victim. Regenerating predictions remains a distinct check from rescoring saved predictions.

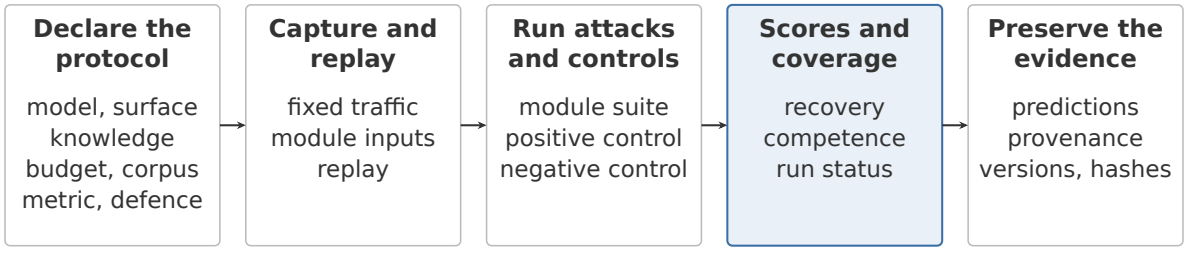


Figure 1: Researchers can check and extend an evaluation from its recorded observations and predictions. Compatible attacks reuse the same messages under stated access and resource limits. Undefended references record attack performance without a defence; other controls test the comparison. Scores and coverage reports distinguish measured outcomes from missing tests. Saved records link each result to its settings and source files.

2.4 Modules, coverage and evidence

Each module declares its supported modes, input requirements, threat metadata and execution function. The registry rejects duplicate names. A run writes configuration, threat model, provenance and results to a common record, retaining completed measurements if execution stops. A new attack can therefore enter a frozen comparison through the same input and output contract, with its own controls and budget.

Three outputs answer different scientific questions. The scorecard evaluates the measured outcomes under the declared rule. The coverage report marks applicable threats as covered, partial, unmeasured or out of scope. The module-validation matrix records each implementation's qualification status as reference conformance, adaptation, real-model qualification, synthetic qualification or unmeasured capability. An execution failure is not a negative attack result, and registration alone is not real-model validation. The complete matrix is Supplementary Table S15.

The evidence procedure checks files against retained hashes, recomputes metrics from predictions and regenerates predictions from pinned inputs. Optional hash-chained journals reveal edits, deletion or reordering relative to retained hashes. An unsigned chain can be recomputed, so it does not authenticate an untrusted producer. Scientific validity instead rests on the protocol, controls and reproducible analysis. Changes to the attack set or scoring policy receive new versions, preserving earlier results for comparison.

These operations serve three research tasks. Attack developers compare a new module on existing captures. Defence researchers vary the target while holding the evaluation plan fixed and measure both leakage and utility. Reviewers verify and re-score the retained evidence before attempting full regeneration.

3 Scoring and Decision Rules

The scoring rule asks how much text the strongest tested attack recovers above a simple guess. For token reconstruction, that guess is the most common token in the attacker's calibration text. The rule compares each attack's upper recovery bound with this baseline, then takes the largest difference among attacks with valid measurements. A defence can pass only with the required repeated runs and a linked undefended reference that meets the control rule below. A pass applies to the tested settings and attacks; the coverage report identifies what remains untested.

Let $\hat{p}_a$ be attack a's recovery over n_a scored tokens, and let $p_{0,a}(C)$ be the baseline's recovery on those tokens. For applicable attacks with valid measurements, the tested attack envelope is

$$L_{\mathrm{run}}(\mathcal{E}) = \max_{a \in \mathcal{A}_{\mathrm{run}}(\mathcal{E})} 100\,[U_{95}(\hat{p}_a, n_a) - p_{0,a}(C)]\,. \tag{2}$$

The bracketed quantity is the attack's *excess*, expressed in percentage points after multiplying by 100. U_{95} is the versioned one-sided Wilson upper bound, using $z = 2.5392$. This multiplier originated in a five-percent error budget across nine known-plaintext probe attempts and is a fixed convention for other attacks. It is not a simultaneous confidence guarantee over all 64 modules. Correlation among tokens also limits a literal binomial interpretation; the gate is an operational screening convention, not a distribution-free privacy guarantee.

A candidate defence passes when its largest excess is at most one percentage point, its evidence includes at least three random-seed runs and a linked undefended reference qualifies under `scorecard-policy.v3`. The reference must cover the same attacker family and contain a result with excess above one percentage point. The software matches report-level references by family; explicit links are declared by the authors and require their check. A missing or non-qualifying control makes a candidate pass `INSUFFICIENT_EVIDENCE`. High measured recovery remains visible regardless of control availability. If no valid attack ran, the envelope is unmeasured. These rules were fixed before defence testing and retain their named versions in the result records.

Even an attack exactly at the baseline has a positive excess at finite sample size because its upper bound exceeds its point estimate. The record reports this *floor*. At small sample sizes it can exceed the one-point threshold, so a failed gate must be read alongside the estimate and interval to distinguish measured leakage from insufficient precision. The control criterion also uses an upper bound: a sufficiently large floor can satisfy it without high point recovery. The control's recovery, baseline and floor require inspection; eligibility alone does not prove competence. With a fixed protocol and existing measurements, adding an attack can only increase the envelope or leave it unchanged. The accompanying coverage report states what remains untested.

This token gate is distinct from the inferential tests used to compare attacks below. Whole-document and span recovery retain their own measures. Membership, timing and agentic outcomes do not pass through the token bound. A verdict names its corpus, attacker set and settings, rather than extending automatically to another protocol.

4 Experimental Design

4.1 Controlled reconstruction

The principal experiment uses GPT-2 124M [17], split after layer three or six, and pinned 32-token WikiText-2 passages [10]. Each run uses 32 public calibration, 128 training and 256 held-out evaluation documents. Ten independently seeded models are fine-tuned with AdamW [9], then frozen and attacked with two attack seeds. The fixed snapshot supplies activations from the client layers and gradients of the next-token loss averaged over all 256 evaluation documents. The public and trained layer conditions are evaluated separately.

The activation-only attacker fits a ridge decoder from public activations to token embeddings, selects 32 candidate tokens per position and uses width-two beam search to match recomputed activations. The joint attacker keeps 16 activation candidates and adds 16 from the negated gradient at the previous position, filling overlaps from the activation list. The first position retains activation candidates. Candidate count and beam width are matched. Shuffling gradients between documents and replacing their candidates with random tokens remove gradient information while preserving the search structure. A backward-only heuristic provides a gradient-only comparator. Full mechanics appear in Supplementary Section S2.

The primary test was fixed in advance. At depth six with public layers, the 95% interval for the joint-minus-activation token gain must lie above two percentage points. The bootstrap resamples trained models and documents independently, using 100,000 replicates and seed 93281. Forty-seven secondary contrasts use Bonferroni-corrected bootstrap intervals. The campaign contains 3,200 training measurements and 480 explicitly reused activation-only inference measurements, giving 3,680 records over the same held-out set. All records and journals re-verify against their manifests.

4.2 Additional protocols and comparators

The corpus study applies a linear inversion decoder without beam search to source code, templated synthetic clinical notes and WikiText-2, using public layers at depth six. Its inference rows pool three seeds and 3,072 scored positions per corpus. Each corpus has its own calibration-fitted baseline.

The defence-aware study optimises continuous token embeddings through the client layers and the actual compressor. It uses the separate pilot WikiText-2 corpus, 128 optimisation steps, three seeds and compressed widths 64, 128 and 256 across all eleven internal depths. The 198 stable-compressor cells use public layers and training traffic. The 99 co-trained cells use trained layers, the final compressor and held-out final-state inference captures. These are separate observation and knowledge protocols.

		Token recovery (%)		Exact documents (%)	
Depth	Layers held	Activation	Joint	Activation	Joint
3	Public	97.39	98.29	43.13	55.51
3	Trained	97.81	98.72	51.41	66.13
6	Public	**94.20**	**97.38**	**13.71**	**37.77**
6	Trained	94.81	98.27	20.23	58.55

Table 1: A modest token gain yields substantially more complete documents. Each entry aggregates 5,120 document-level scores from the same 256 held-out documents across ten trained models and two attack seeds. Highlighting identifies the prespecified primary setting.

We also adapt BiSR [2] to the single split in our harness. Reference conformance compares the port and archived reference in separate processes on fresh held-out documents drawn from the WikiText-103 training split. The subsequent paired comparison reuses the principal campaign's targets and controls, with fresh inverters trained on public auxiliary text. BiSR receives 20 inverter-training epochs and 32 gradient-matching plus 800 activation-matching refinement steps. Each attack retains its own fixed budget. The comparison measures their specified configurations, rather than an equal-compute ranking or the original two-cut BiSR protocol.

5 Results

5.1 Returned gradients increase reconstruction

The primary test passed. At depth six with public client layers, the joint attack recovered 97.38% of tokens against 94.20% for activation-only reconstruction, a gain of 3.17 percentage points, with 95% interval $[2.72, 3.64]$. Exact-document recovery rose from 13.71% to 37.77% (Table 1). All 47 secondary contrasts favoured the joint attack after Bonferroni correction.

The shuffled-gradient and random-candidate controls recovered 93.91% and 93.89%, respectively, compared with 49.98% for the backward-only heuristic. The joint gain therefore depends on the gradient's information, not simply on replacing half the candidates. Gaussian gradient noise at 0.1 times the per-document root mean square and eight-bit quantisation left recovery at 97.31% and 97.33%.

Figure 2 shows the wider range across attack configurations. At the same undefended depth-six boundary, exact-document recovery varies from 1.70% to 58.55%. These configurations differ in observation, knowledge and effort. Table 1 supplies the matched comparison.

Defended comparisons also separate token, document and utility outcomes. Secret mixup, which blends each activation with a private decoy from an earlier batch, left 83.01–90.67% token recovery while limiting exact-document recovery to at most 1.48%. Name-span recovery remained 47.85–66.15% across 716 annotated spans in 249 documents. The annotations were fixed before the run, with AI assistance for names; these are semantic annotations, not human gold labels. Geometric-mean perplexity rose by 31.8% at depth three and 26.9% at depth six. Replica median, an integrity defence that sends activations unchanged, kept token recovery within two-thirds of a point of its undefended comparator. Supplementary Section S3.1 gives the defended protocols and full span results.

5.2 Corpus and defence change the conclusion

The same inversion attack recovered every token of the templated clinical notes, 64.94% of code and 63.25% of WikiText-2 (Table 2). Baselines also differed, from 2.64% to 20.31%. All three undefended conditions failed the token gate. The clinical result characterises regular synthetic templates. It does not measure recovery from clinical records.

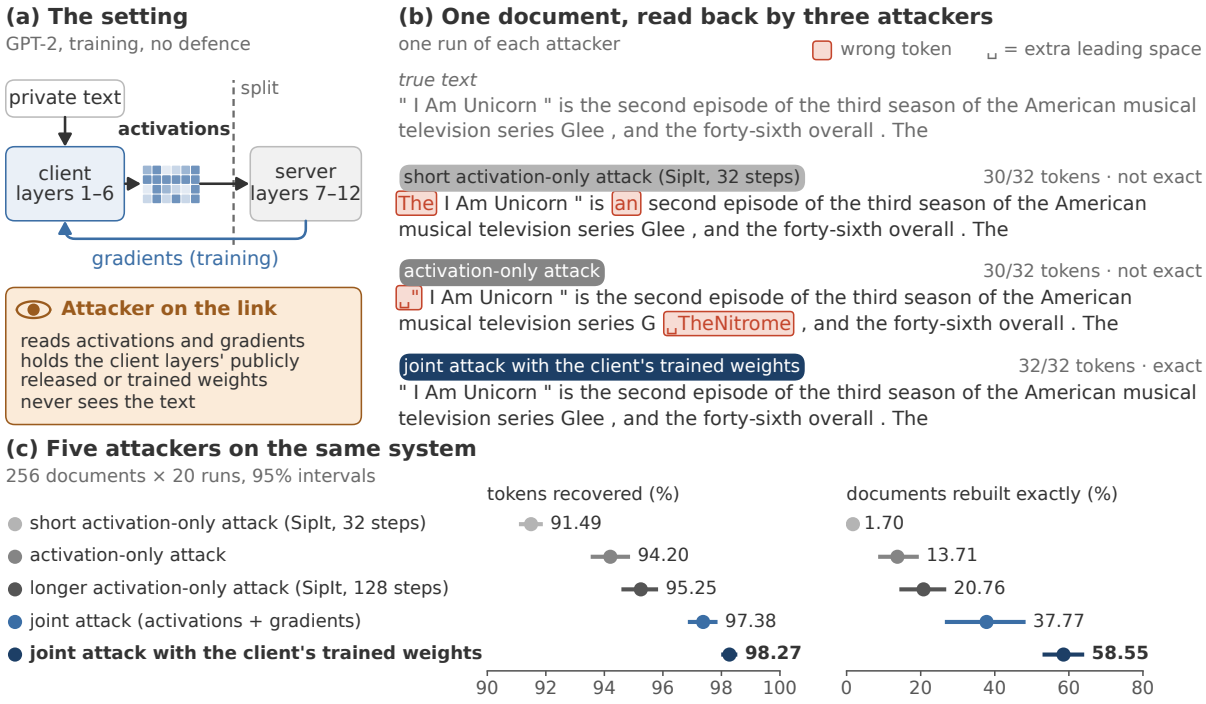


Figure 2: Attack configuration changes how much text is recovered. All configurations use the same depth-six GPT-2 targets and 256 held-out WikiText-2 documents over ten trained models and two attack seeds. (a) Observations at the split. (b) One recovered document, with errors boxed. Among those the strongest attacker recovered exactly in that run, this one is closest to the three attackers' average token recovery. (c) Token and exact-document recovery with 95% intervals from resampling trained models and documents independently. All but the final configuration use public layers, while the final one uses trained layers.

Corpus	Baseline (%)	Recovery (%)	Excess (pp)
Source code	20.31	**64.94**	46.78
Synthetic clinical-style notes	6.25	**100.00**	93.75
WikiText-2	2.64	**63.25**	62.79

Table 2: Corpus choice changes both recovery and its baseline. Undefended GPT-2 at depth six, public layers, inference, three seeds and 3,072 pooled positions per corpus. Excess uses the Wilson upper bound, not the recovery point estimate minus the baseline.

Knowing the defence also matters. The validated white-box path recovered 18.85–68.59% mean tokens across the stable-compressor depth profiles, despite reducing each 768-dimensional activation to 64–256 numbers. Co-trained bottlenecks showed 23.52–67.68% recovery across their separate final-state protocol. Each depth mean averages three widths and three seeds. Twenty checks validated the gradient path, work accounting and capture matching. All co-trained captures replayed exactly from their saved parts. The full depth table and budgets are in Supplementary Section S3.3.

Horizon and longer-context studies extend these settings without merging their protocols. No verdict changed between 20 and 100 training steps in the six-defence pilot battery. A 500-step extension found that a fixed random projection suppressed the declared, projection-unaware inversion attacker below its majority baseline, but also largely prevented useful learning. Cross-entropy remained 7.42–7.75 against 4.39–4.44 without a defence. On 64- and 128-token code and Wikipedia passages and 32-token dialogue, undefended inversion recovery ranged from 76.56% to 90.22%. Gaussian activation noise with $\sigma = 15$ reduced it to 9.96–55.05%. The supplement retains the individual conditions, controls and utility measurements rather than combining them

Depth	Layers held	Activation	SipIt-128	Joint	Refined BiSR
3	Public	97.39	97.45	98.29	96.16
3	Trained	97.81	98.89	98.72	97.79
6	Public	94.20	95.25	**97.38**	**93.21**
6	Trained	94.81	98.14	98.27	98.25

Table 3: No tested configuration wins in every setting. Token recovery (%) on the same 256 documents, ten trained models and two attack seeds, with attack-specific fixed budgets. Activation, SipIt and joint results are reused controls from the main campaign. The inverter-only stage, conformance protocol and complete work counts appear in the supplement.

into one attack-strength score.

5.3 Published-attack conformance and comparison

Our BiSR port and the archived reference produced identical tokens in all 3,072 paired predictions under the single-seam contract. This establishes implementation conformance on those inputs before comparison with other attacks.

In the adapted experiment, refined BiSR recovered 93.21% at depth six with public layers, against 97.38% for the joint attack. The paired difference was -4.17 percentage points, with 95% interval $[-4.79, -3.53]$. With trained layers, the difference was -0.02 points, interval $[-0.63, 0.58]$. Across 15 prespecified secondary contrasts, ten intervals lay below zero, one above and four crossed zero. The configurations in Table 3 show why the framework retains several attacks and their budgets rather than naming a universal winner.

5.4 Validation of the measurement procedures

We selected 25 existing development-test functions before rerunning them from the reviewed framework revision. Parameterisation yielded 37 cases covering recovery metrics, score calculations, control eligibility, observation boundaries, solver fidelity, resampling and evidence validation. The design combines known-answer fixtures, independent calculations, fault injection and metamorphic tests of prescribed relationships between outputs under controlled changes to inputs [19]. Inputs comprised constructed sequences and arrays, small randomly initialised models and synthetic evidence records.

All 37 selected cases matched their expected outcomes, with no failures or skips. Changing text available only to the BiSR scorer left predictions byte-identical. Supplying that text to the predictor was rejected. Removing or weakening a required positive control prevented a candidate pass. In a dependence stress case, increasing perfectly correlated document columns from two to 500 preserved the interval $[0, 1]$ instead of treating copies as independent evidence. Intact evidence was accepted and the specified corruptions rejected. These fixed-case results establish conformance of the selected measurement procedures, complementing reference conformance and real-model experiments. Supplementary Table S11 reports all seven property groups and their case counts.

6 Cumulative Evidence and Scope

Taken together, the programme establishes consequential leakage under several independent protocols. The findings include exact structural leakage in a deployed Qwen split-training system, high text reconstruction in the controlled GPT-2 campaign, and further corpus-, placement-, depth- and defence-dependent failures. The Qwen audit recovered the real/decoy partition on all 4,096 frames in its main run even though the forward-only privacy gate passed [15]. Structural recovery and text reconstruction remain distinct outcomes.

The broader qualification programme covers Qwen3-0.6B, Qwen3-8B, Qwen3.5-27B, Qwen3.5-35B-A3B, Mistral-7B-Instruct and Mistral-NeMo-12B-Instruct in diagnostic, ablation, robustness, obfuscation and agentic roles. These campaigns record expected, negative and privacy-breaking outcomes across multiple models and defences. They establish that modules have been exercised under their declared contracts, alongside the controlled GPT-2 recovery results. Supplementary Table S10 records each protocol's threat model, module, control, model, corpus, budget, outcome, comparator and evidence tier.

The live agentic tier illustrates the importance of retaining negative outcomes. A Qwen3-0.6B planner with reasoning disabled produced measurable actions in all ten fixed settings and no policy violation in twenty trials, but blocked all six sanctioned peer-request trials. This qualifies planner/tool integration and identifies over-blocking. It does not establish general agentic security. Synthetic qualification, reference conformance and target-specific security measurements occupy separate tiers throughout the module matrix.

The strongest quantitative reconstruction comparisons in this paper concern GPT-2 124M under the stated corpora, lengths and budgets. Their main value for the framework is that changing a protocol component changes the conclusion, while retained captures and predictions make that change inspectable. Corpus-specific baselines, utility measurements and matched controls supply the interpretation. Aggregating different protocols into a single success rate would discard it. Longer contexts, new targets and additional attack budgets can extend the evidence through the same procedure.

All experiments used author-controlled systems. Live endpoint modes require an explicitly supplied host, and the agentic planner accepts local model targets only.

7 Related Work

VFLAIR-LLM [7] combines configurable split inference and fine-tuning with attacks and defences. garak [4] organises security probing through generators, probes, detectors and reports, while ML Privacy Meter [12] measures training-data privacy risk through membership inference. LLM-Attacker concentrates on the experimental contract for split and distributed systems, covering explicit observations and knowledge, shared captures, linked controls, execution coverage and versioned evidence. Our comparisons test the consequences of those choices.

Gradient reconstruction includes DLG [21], TAG [3] and LAMP [1], while activation inversion includes Vec2Text [11] and SipIt [13]. BiSR [2] combines a learned inverter with gradient and activation refinement for split fine-tuning. We assess its reference conformance under the local single-seam contract and compare the adaptation with fixed-budget alternatives. The remaining modules are local implementations or adaptations with their evidence tiers stated separately, not claims of reproducing every cited paper's complete protocol.

Earlier split-learning attacks demonstrated inversion, model stealing and label inference [14, 6]. Subsequent work examines why depth alone does not guarantee privacy [5] and recovery during split inference [8]. These motivate evaluation across split depths and observation surfaces. The supplementary catalogue records the contracts and qualification evidence for membership, side-channel, application and agentic modules. Their endpoints remain separate from the text-reconstruction gate.

8 Conclusion

LLM-Attacker makes privacy evaluation an explicit, reusable experiment. Its modules share observation contracts and evidence records, its scoring requires baselines and positive controls, and its coverage report keeps untested threats visible. The experiments establish that attacker knowledge, returned gradients, corpus, defence and attack configuration materially change recov-

ery. Reference conformance and selected measurement-validation tests support the procedures used to measure those effects.

Researchers can use the same framework to add attacks to retained captures, compare defence revisions and give reviewers predictions they can re-score. Every privacy claim for a split model should therefore name its attacker, corpus and scoring rule, report the supporting controls and leave untested attacks labelled as untested.

Evidence and Supplementary Material

The supplementary material appended to this article retains the full protocols, per-condition results, 64-module validation matrix, source identifiers and reproduction guide. Its sections, figures, tables and equations use an S prefix; all cross-references resolve within this document. The canonical code location is `https://github.com/setloop-io/LLM-Attacker`. The retained campaign evidence, measurement-validation records and executable source snapshots are collected in Hugging Face release v1.0. The additional 37-case validation package is at `validation/measurement-20261001/`. The release includes a verified source archive and a Git-independent runner so that the validation can be reproduced without the former repository history. File verification, rescoring and prediction regeneration are documented in Supplementary Section S4. The archived evidence release is available at `https://huggingface.co/datasets/Setloop/The-LLM-Attacker-Unified-Red-Team-Framework-for-Privacy-Evaluation-of-Split-and-Distributed-LLMs/tree/v1.0`. Data files require Hugging Face login and manual approval. The source repository currently requires authorised GitHub access.

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Supplementary Material

The following sections retain the detailed threat models, attack procedures, experimental results, release information and complete 64-module validation matrix behind the main text. They describe the same experiments and artifacts, not an additional independent evidence base. Unprefixed references identify the main text. The retained campaign and measurement-validation artifacts are collected in tag `v1.0`, as specified in Section S4.

S1 Threat Model and Setup

Figure S1 shows the two parties, the traffic between them and where our attacker sits. The *client* owns the text and runs the first layers, from the token and position embeddings through the block at the split depth. The *server* runs the rest: the remaining blocks, the final layer norm and the output projection to vocabulary logits. In training and in inference the client sends the server an activation for each token; during training it also receives back a gradient for each activation, which is the next-token loss differentiated with respect to that activation.

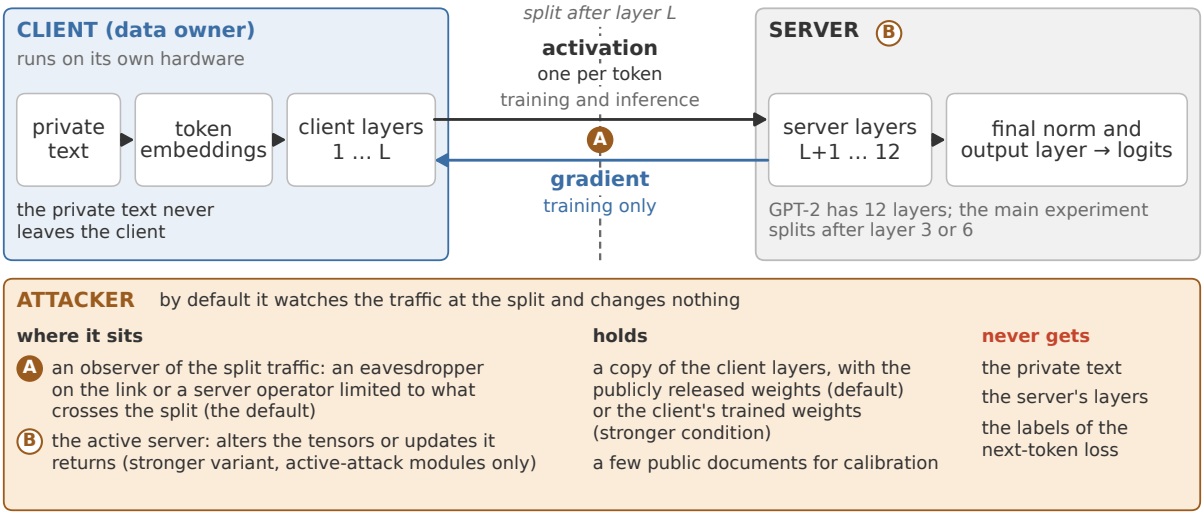


Figure S1: The client keeps the text and its first layers; the attacker reads what crosses the split. Marker A is our attacker, a passive observer of the split traffic. Marker B is the active server, a stronger attacker used only by the active-attack modules.

Our *attacker* watches this traffic at the split and alters none of it. It can be an eavesdropper on the link or an operator of the server who sees only the tensors that cross the split. It records every activation and, in training, every gradient, and it keeps its own copy of the client's layers along with a handful of public documents for calibration. The private text and the server's layers stay out of its reach, and so do the next-token labels, which are the text shifted by one position. For this reason we do not treat the whole server as our attacker, since a server that computed the loss itself would hold those labels. The code that captures the traffic in our experiments runs the server's layers and computes the loss from the labels, yet it passes neither to the attack; we use the text only to score the attacks.

A server that alters the tensors or updates it returns, to make the client leak more, is a separate and stronger attacker, the *active server*; only the active-attack modules take that role.

The attacker's copy of the client's layers is by default the *public copy*, the pretrained model's first layers as publicly released, which anyone can download. As a stronger condition we give it instead the *trained copy*, the client's first layers after training together with the client's fine-tuned token embeddings. The client does not publish these weights; we give them to the attacker to

measure what a leak of them would add, and we count the time to export them in the attack's cost.

Some modules assume a stronger attacker still, and each module states which one it assumes. A *labelled* attacker also holds activations whose tokens it knows, either because it chose that text itself or because some source reveals the true token behind an observed activation; we call such a source an *oracle*. A *label-free* attacker has only public text and public weights. Some defences also multiply every activation by a secret key and replace the key at each *epoch*, one full pass over the training data; we call this *secret rotation*. A *per-epoch* attacker must start again after each new key, while a *cross-epoch* attacker carries information from one key to the next, for example a structure that all keys share or key material leaked in earlier epochs. The key-solving modules model these attackers; the one experiment in this paper that runs them uses no rotation (Section S3.2).

The same modules run from three positions without code changes: installed next to the defence, probing a fleet of machines over a network connection, or as a relay on the client-to-server link. The relay forwards both directions unchanged and copies them into a log; once the log reaches its size limit, it stops copying and keeps forwarding.

The activation and the gradient are the first two of the nine observation points in Figure S2. The other seven are the server's weights, its outputs, leftover GPU memory, caches and timing, the shape of the traffic, reasoning records returned by an API, and the labels and properties of the data. Apart from the shape of the traffic, these points sit on the server or the client, beyond what an observer at the split can see, so the modules that read them assume a stronger attacker. Table S1 lists what crosses each point and the modules that attack it, including five points we added after reviewing the threat model.

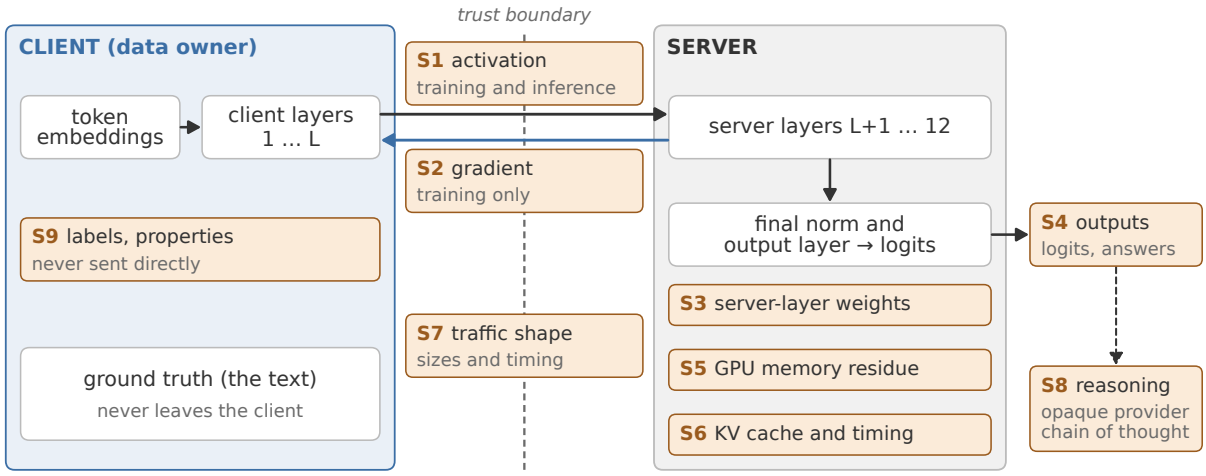


Figure S2: An attacker can observe three surfaces on the link, one on the client and five on the server side. The client holds the private text, the token embeddings and the client layers; the server holds the remaining layers, the final norm and the output layer. Orange boxes are the observation surfaces S1–S9 of the surface model; control-plane metadata, persistent stores and router state extend it. Table S1 lists the attacks on each surface. Cutting the model twice, into three parts, adds a second place where activations cross; mixture-of-experts server layers leave the activation unchanged but add routing decisions as a possible leak.

#	Observation point	What crosses it	Attacks on it
S1	activation (forward)	the activations of each training batch or prompt	`seam-inversion`, `vec2text-inversion`, the key-solving modules for secret rotation, `latent-probe`, `latent-sensitivity`, `wire-eval`
S2	gradient (backward)	the gradients returned for those activations	`gradient-inversion` (DLG++, our multi-restart extension of deep leakage from gradients; iDLG; and a local variant of TAG), `batch-label-recon`, `label-scoring`, joint reconstruction
S3	server-side weights	never (held by server)	`body-extraction` (functional distillation)
S4	outputs / logits / answers	API responses	`output-inversion`, `trace-inversion`
S5	GPU memory residue	freed buffers, local memory	`gpu-memory` (residue, scrub-check)
S6	cache / timing	key-value and prefix caches, contention, acceptance patterns	`kv-timing` (with a speculative-decoding acceptance mode), `gpu-memory` (unified-memory timing mode)
S7	traffic shape	packet sizes/timing on the link	`wire-shape` (via `mitm-proxy` capture)
S8	API reasoning records	encrypted chain-of-thought records	`reasoning-replay`
S9	labels / properties	never directly	`lira-membership`, `membership`, `label-scoring`
Extended observation points, added after our review of the threat model:			
S10	protocol / control plane	negotiation, session binding, epochs, keys	`protocol-integrity`, `key-session-abuse`, `seed-disclosure`
S11	persistent stores / telemetry	captures, logs, checkpoints, crash dumps, backups	`artifact-telemetry-leak`, `remote-fetch`
S12	mixture-of-experts router metadata	expert IDs, queue depth, capacity overflows	`router-side-channel`
S13	AI artefacts / supply chain	models, tokenisers, adapters, datasets, plugins	`supply-chain`, `poisoning-backdoor`
S14	application context	system prompts, retrieved documents, tool schemas, memory	`prompt-injection`, `hidden-context-exposure`, `rag-vector-abuse`

Table S1: Where each attack reads, and which modules read it. What crosses each observation point of Figure S2 and the modules that exploit it.

S2 How the Attacks Work

The strongest reconstruction attack rebuilds a document in four steps. The activation-only attacker uses steps one, two and four; the joint attacker adds step three. This attack is a self-contained routine, distinct from the registered inversion module's own joint variant, which explores candidates differently. The four steps match the numbered badges of Figure S3.

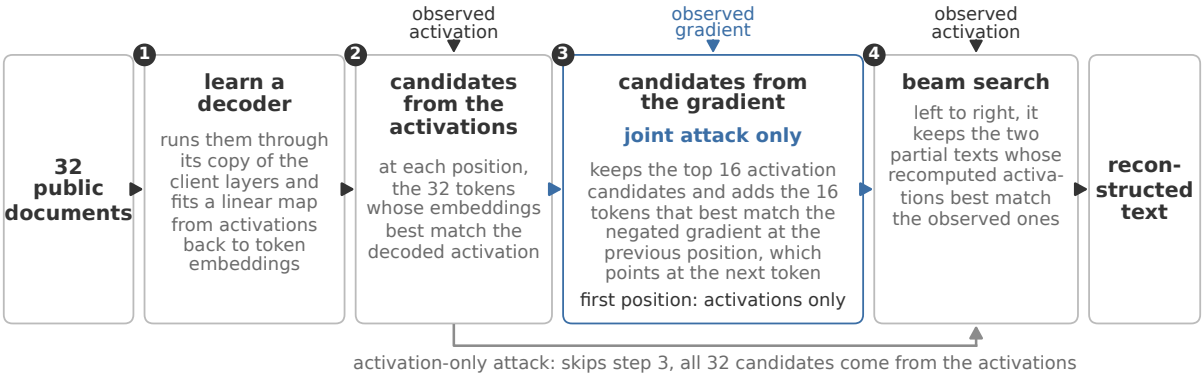


Figure S3: The joint attack takes half of its candidate tokens from the gradient and otherwise runs the same steps as the activation-only attack. The numbered badges correspond to the four stages described below.

1. **Learn a decoder.** The attacker sends 32 public documents through its copy of the client's layers and records, for each token, the activation and the token behind it. From these pairs it fits a linear map, a ridge regression held stable by a penalty on its size, that takes an activation back to a token embedding, the vector the model gives each word in its vocabulary.

2. **Candidates from the activations.** At each position, the attacker applies this map to the observed activation and lists the 32 tokens whose embeddings have the highest *cosine similarity* with the result, meaning they point most nearly in the same direction.

3. **Candidates from the gradient (joint attack only).** The gradient comes from the next-token loss, whose target at each position is the document's next token, so the gradient at one position carries a signal about the token that follows it. The joint attacker keeps the top 16 candidates from the activation and adds the 16 tokens whose embeddings have the highest cosine similarity with the negated gradient at the previous position, filling any overlap from the activation list. No earlier position predicts the first token, so the first position keeps its activation candidates.

4. **Beam search.** A width-two search reads the text left to right, extending its two leading guesses with each candidate, pushing them back through the client's layers, and retaining the two whose recomputed activations lie closest to what crossed the split. The two attackers try equally many partial texts and part ways only on their candidate lists.

In the main experiment the attacker reads one fixed snapshot of this traffic rather than live training traffic. We fine-tune the whole model, freeze it, split it, and pass the 256 evaluation documents through it once. The activations come from the frozen client layers, and the gradients from the next-token loss averaged over all 256 documents.

The *shuffled-gradient* control strips the gradient of its information by giving the joint attacker another document's gradient in place of the right one. The *random-candidate* control retains the activation half of the list and draws the rest uniformly from the vocabulary, so no gradient enters. The *backward-only* heuristic uses the gradient alone: at each position it picks the token with the highest cosine similarity to the negated gradient at the previous position, with no activations and no beam search. We also run SipIt [13], which rebuilds the input one position at a time: for each position it searches for the token whose activation, given the prefix already recovered, matches the observed one, with a budget of steps per position. We implemented the published algorithm ourselves; when the budget runs out it returns the best token it has tried, so its guarantee of exact recovery no longer holds.

The experiments also use a *white-box* attack, which knows the defence and searches for continuous token embeddings that, after passing through the client's layers and the defence, reproduce the observed message, then reads off the nearest token at each position. The *inversion attack*, a form of vanilla model inversion, fits the same linear map from activations to embeddings as step one, on its own calibration text passed through the public layers, and reads off the nearest token without a beam search. *BiSR*, or bidirectional semi-white-box reconstruction [2], works in two stages: a learned inverter first guesses the text from the forward activations, and an optimisation stage then refines the guess. In our port, the inverter (which the code names SIP) is a small recurrent network trained on public text passed through the public layers, and the refinement (which the code names bf) first matches the observed gradient and then matches the observed activation. Table S2 lists each signal, when it crosses, how an attack uses it and what it reveals.

Signal	When it crosses	How the attack uses it	What it reveals
Activation, one vector per token	Training and inference	The learned decoder turns each activation into a short list of likely tokens; a beam search keeps the guesses whose recomputed activations match. SipIt and the inversion attack also read it.	Token, whole-document and span recovery
Gradient of the next-token loss	Training only	Each gradient points at the token that follows it, so it supplies half of the candidates at each position.	What the gradient adds to the same scores
Size of the returned gradient, per document	Training	The summed gradient size scores whether a document contains a given token.	Properties of the text, as an area under the ROC curve
Controls: another document's gradient; random tokens in place of the gradient's suggestions	Training	The same attack, with the gradient's information removed	Whether the gain comes from the gradient

Table S2: What an observer at the split reads, and what each signal gives up. The controls replace what the gradient carries and keep the number of candidates, which shows that the gain comes from the gradient's information.

S3 Detailed Experimental Results

Every experiment below is tied to a released machine-readable result and a hash-checked record. We report each experiment against its prespecified threat model, controls and statistical procedure, then compare the conclusions across observation grants, corpora, attack paths and published attacks.

S3.1 What the attacker observes

The main experiment fixes its plan before running. It uses 256 held-out documents per run and ten client models trained with different random seeds, each attacked twice with different seeds for the attacks' own random choices. The primary statistical test fixed in advance is the *two-point gain test*: at depth six with the public copy of the layers, the joint attacker's token recovery minus the activation-only attacker's must have the lower end of its 95% interval above

two percentage points. The interval comes from a bootstrap that resamples trained models and documents independently. The protocol also prespecified 47 secondary comparisons, each tested with a Bonferroni-corrected bootstrap interval. The client's model is GPT-2 [17] (124M) on pinned 32-token WikiText-2 passages [10], split after the third or the sixth layer, with 32 public calibration, 128 training and 256 evaluation documents. Training uses AdamW [9]. The attacker holds either the public first layers or the client's trained first layers. It attacks the fixed snapshot described in Section S2: in training mode it reads the activation and the gradient, in inference mode the activation alone. In all, 40 attack configurations (28 without a defence, 12 with one) run against ten trained models at both depths with both copies of the layers, for 3,200 measurements; the six activation-only configurations are reused for inference, adding 480, for 3,680 in total, each on the same 256 documents. Sixteen slices ran on two machines with hash-chained logs; the sharding, the merge and the counts are in Section S4.

The gain test passed (Table S3). At depth six with the public layers, the joint attacker recovered 97.38% of tokens against 94.20% for the activation-only attacker, 3.17 points more (95% interval $[+2.72, +3.64]$). The 47 secondary comparisons cover the joint attacker against the activation-only attacker, the shuffled-gradient control, the random-candidate control and the backward-only heuristic, at both depths, with both copies of the layers and with candidate lists of 8, 16 and 32 tokens, leaving out the gain test itself. In all 47 the joint attacker came out ahead, with intervals above zero after a Bonferroni correction for the number of comparisons. The backward-only heuristic, which reads the gradient alone, recovers 49.98% of tokens in the same setting. In the two controls, the shuffled gradient recovers 93.91% and the random candidates 93.89%, both about 0.3 points below the 94.20% activation-only result, while replacing the true gradient with a perturbed one leaves recovery near the true-gradient result, at 97.31% when we add Gaussian noise whose standard deviation is 0.1 times the root mean square of each document's gradient, and 97.33% when we quantise the gradient to eight bits. Together, these controls isolate the information contributed by the returned gradient.

		Token accuracy		**Whole-document recovery**	
Depth	**Layers held**	**Activation-only**	**Joint**	**Activation-only**	**Joint**
L=3	public	97.39%	98.29%	43.13%	55.51%
L=3	trained	97.81%	98.72%	51.41%	66.13%
L=6	public	**94.20%**	**97.38%**	**13.71%**	**37.77%**
L=6	trained	94.81%	98.27%	20.23%	58.55%

Table S3: The gradient adds little per token but much per document. Token accuracy and whole-document recovery for the two attackers at each depth and copy of the layers. Each entry pools 5,120 document-level scores across the 256 held-out documents. The same experiment supplies the companion paper's [16] table; both draw on this one evidence base.

We also ran paired comparisons against two defences, secret mixup and replica median. Each defended model is trained separately with its defence in place. *Secret mixup* blends each activation with a real activation from an earlier training batch, three parts true to one part decoy. The client keeps secret which earlier activation it mixed in, and it corrects the returned gradient for the decoy. Under it, token recovery ranged from 83.01% to 90.67% and whole-document recovery reached at most 1.48%, while separate utility measurements showed the geometric-mean perplexity rise by 31.8% at depth three and 26.9% at depth six. We also scored annotated *spans*, the stretches of text that hold a name, a number or an identifier. Under secret mixup the attacker recovered 47.85–66.15% of name spans (716 spans in 249 documents), 93.94–96.50% of number spans (297 spans in 142 documents) and 90–100% of identifier spans (two spans in two documents), against 67.36–97.44% of name spans without the defence. The annotations were fixed before the run, with names marked with AI assistance. These semantic annotations are not human gold labels.

Replica median runs three copies of the server's layers and takes the median of their outputs, number by number. It protects the integrity of the server's result and sends the activation unchanged, so token recovery stayed within two-thirds of a point of the undefended attack. Whole-document recovery still moved by up to 7.7 points in the most affected setting (public copy, depth three, SipIt at 128 steps, from 43.77% to 36.11%). The defended model is a separate training run, with three slightly perturbed server copies, so its client layers end slightly different, and because one wrong token spoils a document, small changes per token move many whole documents.

The prespecified contrast and all secondary intervals are recorded in `v3-campaign/review/contrasts.json`; per-arm token, document and span results are in `v3-campaign/review/descriptive_summary.json`, and the merged-run integrity check is in `v3-campaign/merge-verification.json`.

S3.2 Which text the attacker rebuilds

The same attacker recovers very different amounts on different corpora (Table S4). We ran the undefended inversion attack on three pinned corpora, each scored against its own trivial guess: this repository's own Python source code, synthetic clinical-style notes written from templates, and a WikiText-2 reference. The trivial guess itself differs: it scores 2.64% on Wikipedia and 20.31% on code, almost eight times higher. Recovery tracks how regular the text is. The attacker recovers every token of the templated clinical notes, 64.94% of the code and 63.25% of the Wikipedia text, so corpus choice alone moves recovery by up to 37 points. Against the corpus-specific guess rates, excess recovery is +46.78 points for code, +93.75 points for the clinical notes and +62.79 points for Wikipedia.

Finite-sample floors are 2.3–3.9 points for the 768-position training rows (Table S5) and 0.84–1.90 points for the 3,072-position inference rows. Every undefended result exceeds its corpus-specific guess by at least 46 points, so every undefended condition fails the privacy gate. Added Gaussian noise (σ=0.1) changed mean recovery by at most 1.4 points; the paired intervals include zero for code and Wikipedia and are exactly zero for the templated clinical corpus.

Corpus	Trivial guess	Recovery	Excess	Verdict
Source code (own repository)	20.31%	**64.94%**	+46.78 pp	**FAIL**
Synthetic clinical-style	6.25%	**100.00%**	+93.75 pp	**FAIL**
WikiText-2 reference	2.64%	**63.25%**	+62.79 pp	**FAIL**

Table S4: Corpus choice moves recovery by up to 37 points. Undefended inversion (GPT-2, depth six, inference rows, n=3,072). Each corpus is pinned and scored against its own trivial guess; the floor for these rows is 0.84–1.90 points. Training rows (n=768) show the same direction; their guess rates and floors are in Table S5. Both row counts pool three seeds.

Corpus	Guess count	Guess rate	Floor (pp)
Source code	154/768	20.05%	3.91
Synthetic clinical-style	48/768	6.25%	2.60
WikiText-2 reference	36/768	4.69%	2.34

Table S5: Guess rates and floors for the training rows of the domain battery. Each row pools 768 token positions across three runs. The floor is the excess an attack scores at the guess rate; full-precision constants are in Section S6.

The machine-readable domain results are `domain-shift-v1/domain-shift-v1-summary.json` and `domain-shift-v1/domain-shift-v1-defended-summary.json`.

The evaluator also chooses how long the client trains. A long-horizon battery ran six defences at 20 and 100 training steps, over three training seeds, on the *pilot corpus*: 96 pinned WikiText-2 passages from the dataset's training split, which we used while building the framework and which are separate from the main experiment's held-out documents (Table S6). The defences are lightweight stand-ins for defences in the VFLAIR-LLM collection [7], each a plain transform. *Added noise* adds Gaussian noise with standard deviation 15 to every number of the activation. *Sparsify-10%* keeps only the largest tenth of each activation's numbers. *Mid-lite*, named after the mutual-information defence but learning nothing, projects the activation through a fixed random map down to 64 numbers. *Santext*, named after the SanText text-sanitisation defence, swaps each input token, with probability 0.3, for one of its nearest tokens in the embedding table. *At-lite*, named after adversarial training but performing none, shifts every activation by a fixed amount along the direction in which activations vary most. The corpus's majority token does not occur in the evaluated passages, giving the battery a fixed 0% guess rate.

No verdict changed between 20 and 100 steps. Against the declared inversion attacker, which does not know the projection, mid-lite remained at 0% recovery. The 500-step extension shows that mid-lite achieves this result by largely preventing learning: mean training cross-entropy is 7.42–7.75 versus 4.39–4.44 without a defence. Added-noise recovery rose from 0.3% in steps 0–7 to 4.2% in steps 50–99, and sparsify-10% rose from 17.5% to 25.9%. A document-clustered re-analysis changed no verdict, although the 20-step added-noise interval ($[+0.47, +1.35]$ points) crossed the one-point threshold.

Defence	20-step excess (pp)	100-step excess (pp)
none	+63.02	+58.89
added noise (σ=15)	+2.42	+3.69
sparsify-10%	+23.52	+25.03
mid-lite k=64	+1.00	+0.20 **(PASS, at floor)**
santext-30%	+46.15	+42.67
at-lite ε=15	+63.17	+59.00

Table S6: No verdict moves between 20 and 100 training steps. Long-horizon defence battery on GPT-2 at depth six (inversion attack, pilot corpus, mean per-run excess over three training seeds, this battery's own trivial guess). Mid-lite passes at the floor at both horizons; every other defence fails.

A 500-step extension ran six trajectories at depth six, with no defence and with mid-lite, over three training seeds, and two machines reproduced the saved-prediction analysis and verified every log. The inversion attack recovered 81.22–81.73% of tokens with no defence and 0.04–0.85% with mid-lite, below its 4.83–4.96% majority baseline. This extends the same declared setting to 500 steps and confirms the verdict across three seeds and two independently reproduced analyses. The horizon, clustered and 500-step records are, respectively, `long-horizon-v1/report.json`, `clustered-gate-v1/clustered-gate-v1-summary.json` and `defence-training-500-step-extension/REPRODUCTION.json`.

S3.3 What the validated white-box attack recovers

The white-box analysis optimises the complete differentiable path from token embeddings through the client's layers and the true compressor to the observed message. Twenty PyTorch checks validate the gradient path, work accounting and capture matching. We evaluated 198 stable-compressor cells across all eleven internal GPT-2 depths and a separate 99-cell co-trained tier on held-out final-state inference captures. Each defence compresses the activation. *PCA compression* sends only the activation's top 64, 128 or 256 principal components out of its 768 numbers (the *width*), fitted once on calibration text. A *trained bottleneck* starts from the same projection and adds a small learned correction, trained to defeat probes that try to read the

tokens. Across three training seeds, both methods and three widths, the attack recovers between 18.85% and 68.59% of tokens on average at every depth (Table S7). Each mean averages the three widths.

Depth	PCA (fixed)	Bottleneck (frozen)	Bottleneck (co-trained)
1	59.16%	60.36%	61.14%
2	66.86%	**68.59%**	67.68%
3	58.32%	58.00%	60.24%
4	56.88%	57.35%	60.22%
5	55.90%	57.02%	56.80%
6	50.89%	52.16%	53.09%
7	48.94%	49.73%	48.01%
8	41.66%	42.05%	41.10%
9	38.82%	37.96%	38.24%
10	31.01%	31.80%	30.31%
11	**18.85%**	19.67%	23.52%

Table S7: The validated white-box attack recovers 19–69% of tokens at every internal depth. Token recovery (%), mean over three training seeds and three widths at each depth. In the first two columns the compressor stays fixed while the client trains (the bottleneck is trained beforehand and then frozen), and the attacker reads training-time traffic with the public copy of the client layers and the true compressor. In the last column the bottleneck trains alongside the client, and the attacker reads inference traffic from the final model, holding the client's trained layers and the final compressor. The stable and co-trained columns report their separate, fully specified protocols.

A separate 99-run tier trains the bottleneck alongside the client, then attacks held-out inference captures from the final state, giving the attacker the client's trained layers and the trained compressor; every capture replays exactly from the saved parts. Like the rest of the sweep, it uses the pilot corpus. We report the stable and co-trained tiers separately, with their verified cells, means and work counts in `all-boundary-whitebox-reconstruction/full/summary.json`; full compute counts and wall times are in Section S6.

We also ran the inversion attack on longer texts and on dialogue (Table S8), at depth six over three training seeds with no defence or added noise, on code and Wikipedia text at 64 and 128 tokens and on 32-token dialogue prefixes from 96 dialogues. Undefended recovery ranges from 76.56% to 90.22%; added noise lowers it to 9.96–55.05%. Each condition uses three training seeds, and the released records include its majority-guess rate. Noise-aware whitening lowers inference recovery from 21.65% to 13.09% and raises training recovery from 11.46% to 13.67%, so the plain and whitening variants expose different signals.

		Training		Inference	
Corpus	Tokens	None	Added noise	None	Added noise
Code	64	78.32%	21.55%	79.70%	26.27%
Code	128	87.37%	23.11%	86.71%	32.92%
Wikipedia reference	64	81.77%	9.96%	82.50%	20.82%
Wikipedia reference	128	88.96%	14.55%	90.22%	26.17%
Dialogue	32	76.56%	36.72%	76.99%	55.05%

Table S8: Longer texts and dialogue leak without a defence; added noise lowers recovery but does not stop it. Inversion-attack recovery (%), means over three training seeds. Added noise is σ=15. Training and inference read different signals.

The verified raw records and their manifests are packaged under the Phase 2 evidence README.

S3.4 Which published attack the evaluator runs

We first checked that our port of BiSR matches the reference. Using the same saved models, inverters, observations and budgets as an archived copy of the upstream source, and running our port and the reference in separate processes on 256 fresh 32-token documents at depths three and six, with inverters trained under three seeds, both produced identical predicted tokens in all 3,072 paired runs; every seed-level difference was zero. These documents, and the 112 auxiliary articles on which the inverters were trained, come from the pinned WikiText-103 training split, with the evaluation documents held out from it; none comes from the official test split, and the selection excludes the inputs of earlier runs and the WikiText-2 splits. This establishes conformance for the single-seam setting used by our harness. BiSR's published *two-cut setting* is a distinct protocol: it cuts GPT-2-large into three parts and exposes activations at one cut and gradients at the other, whereas both signals cross the one cut in our harness.

We then compared the port with our joint attack under a plan fixed before running (Table S9). It kept the main experiment's 256 documents, ten trained models, both depths and both copies of the layers, reused the earlier results as controls, and trained fresh inverters on the public auxiliary text. Each attack used its own fixed protocol budget. At depth six with the public layers, the port recovered 93.21% of tokens against 97.38% for the joint attack, 4.17 points fewer (95% interval $[-4.79, -3.53]$), with the port lower in every trained model, and the whole interval lies more than two points below zero. Across fifteen prespecified secondary contrasts, ten intervals lie below zero, one above and four cross zero. Notably, the refined port matches the joint attack at depth six with trained layers (-0.02 points, interval $[-0.63, +0.58]$), so no attack dominates every setting.

Depth	Layers held	Activation-only	SipIt-128	Joint	Inverter	Refined
3	Public	97.39	97.45	98.29	91.90	96.16
3	Trained	97.81	98.89	98.72	91.90	97.79
6	Public	94.20	95.25	**97.38**	91.67	**93.21**
6	Trained	94.81	98.14	98.27	91.67	98.25

Table S9: No attack wins everywhere. Token accuracy (%) for the adapted comparison. Each row averages two attack seeds, ten trained models and the same 256 documents. Activation-only, SipIt-128 and joint are the reused controls; the inverter (SIP) and its refinement (bf) are the adapted attack. The paired intervals and attack-specific budgets are in the text and supplement.

The exact conformance verdict is `bisr-conformance-validation/RESULT.json`; the adapted comparison and its prespecified intervals are in `bisr-adapted-comparison/analysis/result.json` and `bisr-adapted-comparison/analysis/contrasts.json`.

S3.5 What strength is established

Attack strength depends on the observation and knowledge grants, corpus, budget, target and control. Table S10 separates the shared confirmatory GPT-2 campaign from independent audits and qualification studies. Together they establish structural leakage, corpus sensitivity, implementation fidelity, published-attack conformance and module behaviour.

Table S10: What attack strength and module qualification are established. Rows retain their own threat model, corpus, budget and comparator. The first row is the paper's confirmatory reconstruction result; the remaining rows are independent audits, controlled real-model studies, implementation conformance and multi-model qualification experiments.

Threat model	Module or arm	Positive / negative control	Model	Corpus	Budget	Observed recovery or outcome	Comparator	Evidence tier
Passive observer; public or trained client layers	Joint reconstruction, v3 arm	Shuffled-gradient, random-candidate and backward-only controls	GPT-2 124M	256 held-out 32-token WikiText-2 documents	10 victim seeds × 2 attack seeds; K=32; beam 2	**97.38%** tokens and **37.77%** exact documents with public layers; 58.55% exact with trained layers	Activation-only: 94.20% tokens, 13.71% exact documents	Confirmatory; shared Paper A/B evidence
Compromised cloud observes forward and returned-gradient channels	`latent-probe` plus exact gradient-support read	Injected-leak calibration; shuffled-label negative control	Qwen3-0.6B	Fixed-width WikiText-2 blocks	9 independent seeds; 4,096 frames per main run	Real/decoy partition exact on **4,096/4,096** frames; frequent-token effect about +0.92 pp	Forward-only gate passed; shuffled controls at floor	Independent real-system audit [15]
Passive seam observer; public client layers	`seam-inversion`	Corpus-specific majority guess; undefended positive target	GPT-2 124M	Source code, templated clinical notes, WikiText-2	3 seeds; n=3,072 inference positions per corpus	**64.94%**, **100.00%**, **63.25%**	Guess rates 20.31%, 6.25%, 2.64%	Controlled real-model qualification
White-box observer knows the compressor	`seam-inversion` white-box path	Twenty gradient-path and capture-matching checks	GPT-2 124M	Pinned pilot WikiText-2	198 stable and 99 co-trained runs; 128 optimisation steps	**18.85–68.59%** mean token recovery over every internal depth	Stable and co-trained tiers use fully specified protocols	Controlled real-model qualification
Single-seam observer with BiSR grants	`bisr-baseline`	Archived reference implementation on identical fresh inputs	GPT-2 124M	Fresh WikiText-103 for conformance; main held-out set for comparison	3,072 paired predictions; 20 inverter epochs; 32 backward + 800 forward refinement steps	Reference agreement on **3,072/3,072**; adapted attack 93.21% at depth six/public	Joint arm 97.38%; attack-specific fixed budgets	Reproduced conformance + adapted comparison
Passive seam observer across training horizons	`seam-inversion`	Undefended positive target; majority control; mid-lite protected contrast	GPT-2 124M	Pinned pilot WikiText-2	20/100 steps, 3 seeds	Undefended excess +63.02/ + 58.89 pp; mid-lite +1.00/ + 0.20 pp	Same attacker on undefended and protected captures	Controlled horizon qualification

continued overleaf

Table S10: What attack strength and module qualification are established (continued).

Threat model	Module or arm	Positive / negative control	Model	Corpus	Budget	Observed recovery or outcome	Comparator	Evidence tier
Observer of an obfuscated inference wire	`wire-eval`	Unobfuscated baseline and shared-key identity-restoration arm	Mistral-7B-Instruct	Pinned E9 session inputs	3 paired deployment sessions	Baseline 50% top-1; obfuscated wire **0%**; held-out loss unchanged within precision	Shared-key arm restores identity	Multi-model defence qualification
Passive activation observer	SipIt qualification arm	Undefended target; bf16/fp32 precision twins	Mistral-7B-Instruct	10 victim documents per depth and precision	Depths 1/4/8; bf16/fp32; effective n=10 per cell	**77.03–84.53%** mean token recovery	Precision twins differ by at most 1.41 points	Multi-model attack qualification
Known-pair, stale-key and sharding attackers	`known-prefix`, `stale-key`, `sharded`	Label-free band; fresh-key solve; static and staggered arms	Qwen3-0.6B, Qwen3.5-27B, Mistral-NeMo-12B	Pinned E8/E-R inputs	Measured K grids; repeated solve seeds and sessions	Known-prefix $K_{90} \approx 232$–240; stale key 0.94–2.45% vs fresh 61.20–65.31%; sharding 54.95–67.86%	Label-free band and fresh/static controls	Multi-model attack qualification

Table S10 distinguishes controlled recovery measurements from multi-model qualification and states the threat model, budget, control and measured outcome for every protocol.

Cumulative evidence. Taken together, the programme supports the following statement: LLM-Attacker has exposed consequential leakage under several independent protocols: exact structural leakage in a deployed Qwen split-training system, high text reconstruction in a controlled GPT-2 campaign, and additional corpus-, placement-, depth- and defence-dependent failures. Its broader Qwen and Mistral campaigns provide qualification evidence that different modules have been exercised against multiple models and defences, including expected, negative and privacy-breaking outcomes.

Table S15 accounts for all 64 modules and records the evidence tier, controls and status of each.

S3.6 Live agentic qualification

A live tier used Qwen3-0.6B [20] as the local planner under the same sandbox, permission checks, side-effect logging and budgets as the synthetic tier. With reasoning mode disabled, all ten prespecified settings produced measurable actions. No policy violation was recorded in the twenty fixed scenario trials. All three sanctioned peer-request settings were blocked in both trials, giving six false blocks; the result therefore establishes successful planner/tool integration and identifies systematic over-blocking of sanctioned requests. The full measurements and planner traces are in `agentic-nonthinking-decoding/raw/FOLLOWUP.json` and `agentic-nonthinking-decoding/raw/planner-traces.jsonl`.

S3.7 Framework operation across protocols

The completed campaigns exercise the workflow in Figure 1 at several stages. The main campaign reused captures across attack configurations and assembled 3,680 measurements, including 480 explicitly reused inference measurements, from sixteen shards. Its merge-verification record and release and re-verification guide document the assembly and provenance checks. The BiSR conformance study tests implementation fidelity against a reference executable; its adapted comparison demonstrates how another attack can join a frozen experiment. The white-box study exercises defence-aware reconstruction across internal split depths, and the live agentic study connects a local planner to the sandbox's action and policy records.

The Qwen and Mistral campaigns extend module qualification across the models and defences recorded in Table S15. Together, these studies establish reuse of the framework across reconstruction, robustness, obfuscation and agentic protocols. Their shared methodological role is to connect an explicit attacker contract to controls, measured outcomes and retained evidence. This is the basis for adding a new attack or target to the programme while preserving the interpretation of earlier results.

S3.8 Validation of the evaluation methodology

We tested whether the evaluation code preserves the intended meaning of recovery measurements and rejects specified errors in their supporting evidence. The design combines known-answer fixtures, independent numerical calculations and fault injection with metamorphic tests: checks of prescribed relationships between outputs under controlled changes to the inputs [19]. For example, changing text available only to the scorer must leave the attack's predictions unchanged, while removing a required positive control must prevent a candidate pass.

We fixed a selection of 25 existing development-test functions before re-executing them from the reviewed framework revision on CPU, retaining their original assertions and numerical tolerances. Parameterised inputs expanded the selection to 37 cases across seven measurement

properties (Table S11). The inputs were constructed token sequences, numerical arrays, small randomly initialised models and synthetic evidence records.

Property	Controlled intervention	Expected response	Cases
MV1: recovery metrics	Known token errors, absent spans and misplaced matching text	Exact token and span counts; missing spans stay missing; misplaced matches score zero	**3/3**
MV2: score conformance	Independent Wilson calculations and threshold crossings	Numerical agreement within tolerance; prescribed boundary verdicts	**4/4**
MV3: control eligibility	Retain, remove, weaken or make a linked control unreadable	Backed candidate pass retained; other candidates marked insufficient evidence	**4/4**
MV4: observation boundary	Change scorer-only truth; supply forbidden truth or malformed observations	Byte-identical predictions after truth change; invalid inputs rejected	**2/2**
MV5: solver fidelity	Independent GRU and AdamW calculations; white-box head checks	Outputs and derivatives agree; correct layer prefixes and nonzero embedding gradients	**3/3**
MV6: resampling	Constant data, correlated document duplication and invalid matrices	Known intervals retained in the fixed cases; invalid matrices rejected	**6/6**
MV7: evidence validation	Two intact cases, ten invalid report states and three retained-hash corruptions	Intact evidence accepted; all specified faults rejected	**15/15**

Table S11: All 37 selected validation cases matched their specified outcomes. The last column counts matched outcomes over executed cases, including parameterised inputs; rejecting an invalid input is a successful check. MV1–MV7 identify the case groups in the measurement-validation supplement described in Section S4.

There were no failures or skips. In the dependence stress case, increasing perfectly correlated document columns from two to 500 preserved the interval $[0, 1]$ rather than treating the copies as independent evidence. The BiSR prediction process produced byte-identical outputs after its scorer-only truth file changed and rejected a manifest that supplied that truth to the predictor. These results establish conformance of the selected measurement procedures on their fixed cases. They complement the external-reference comparison in Section S3.4 and the real-model leakage measurements.

S4 Release and Reproduction

The main experiment ran as sixteen deterministic slices across two machines, merged with hash verification, so that depth is never tied to one host; the merge checks that every prediction record and every log verifies and that the runs together equal the planned set. Its 860 binary-tier references are hash-verified against the pinned baseline release, with no missing or mismatched files. The Hugging Face dataset at `https://huggingface.co/datasets/Setloop/The-LLM-Attacker-Unified-Red-Team-Framework-for-Privacy-Evaluation-of-Split-and-Distributed-LLMs` holds the predictions, logs and captures of the main experiment, the domain, horizon and clustered analyses, the two distinct agentic packages, the positive controls, and the code, protocols and inputs. Tag `v1.0`

collects the retained campaign evidence, the final manuscript and the measurement-validation records identified below. Experimental files retain their original hashes; source snapshots preserve the code needed for reproduction independently of the former Git history.

The manuscript, reviewed 64-module taxonomy and released code snapshot have distinct roles; Table S13 records each identifier and scope. The GitHub remote `https://github.com/setloop-io/LLM-Attacker` is the canonical source location, and the tagged Hugging Face release is the reproducibility artifact. The released command-line tool offers self-test, module listing, scorecard, coverage and profile reports, and the synthetic tier needs no deep-learning stack. Tables S12 and S13 give the principal release locations and identifiers.

Evidence	Path in the tagged dataset release
Release root and manuscript	`v1.0`; `paper/`
Main campaign and controls	`v3-campaign/`; `binary-tier/`; `positive-controls/`
Inputs and protocols	`corpora/`; `protocols/`
Domain and horizon analyses	`domain-shift-v1/`; `long-horizon-v1/`; `clustered-gate-v1/`
Validated white-box results	`all-boundary-whitebox-reconstruction/original36/`; `all-boundary-whitebox-reconstruction/full/` (31 files); `phase2-raw-evidence/` (four files, 7.15 GB)
BiSR and adapted comparison	`bisr-conformance-validation/`; `bisr-conformance-run-artifacts/`; `bisr-adapted-comparison/`
Training extension	`defence-training-500-step-extension/`
Agentic evidence	live planner: `agentic-live-v1/`; non-thinking amendment: `agentic-nonthinking-decoding/`
Indexes and checksums	`INDEX.md`; `REPRODUCE.md`; `MANIFEST.sha256`; `BINARY-VERIFICATION.json`; `v3-campaign/EVIDENCE-RELEASE.md`

Table S12: Principal release locations. Current paths in the cleaned `v1.0` tree. Use the tagged paths; former Git commit and release URLs are not navigation targets for this release.

Item	Identifier or scope
Combined manuscript	`paper/`; the concise paper and full supplement are supplied as one PDF and a self-contained arXiv source ZIP
Reviewed framework taxonomy	`source-snapshots/reviewed-framework-64.tar.gz`; 64 registered modules; historical source identifier `6c071a68ae5134522d0a4cd3d7d0fa67acabdfa8`
Dataset code snapshot	`code/` and `source-snapshots/base-framework-code.tar.gz`; historical base source identifier `a3eb53fca5d6395da4c4377e9d8787676c66ae6a`, supplemented by experiment-specific source archives
Maintained framework	GitHub tag `v1.0` contains 65 registered modules; the additional `secure-compute-audit` module is outside the paper's reviewed 64-module qualification matrix
Models and corpus	GPT-2 124M `openai-community/gpt2`, revision `607a30d783dfa663caf39e06633721c8d4cfcd7e`; Qwen3-0.6B `Qwen/Qwen3-0.6B`, revision `c1899de289a04d12100db370d81485cdf75e47ca`; released `corpora/` and `protocols/` are the binding text inputs
Seeds	main experiment: ten trained-model seeds and two attack seeds; white-box and horizon: 11, 22, 33; BiSR check: 31415, 27182, 16180; adapted comparison: attack seeds 17, 29; bootstrap: 93281
Licence metadata	Framework and release packaging: MIT, with `LICENSE` supplied. Third-party components retain their own licence terms

Table S13: Revision, model and access identifiers. The distinct source components are identified separately so that the manuscript, reviewed taxonomy and released code snapshot can be reproduced without relying on old Git refs.

Measurement-validation supplement. The additional validation run in Section S3.8 is supplied in `validation/measurement-20261001/` within release `v1.0`. The corresponding path in the maintained source repository is `validation/measurement-20261001/`. Its original selection, expected outcomes and per-case observations remain in `selection.json` and `run/results.json`; the original `MANIFEST.sha256` is preserved unchanged. The package also contains `source-snapshot.tar.gz`, which matches all 123 retained source-file hashes, and `reproduce_from_snapshot.py`, which reproduces the frozen selection without Git history. Added packaging files have separate digests in `SNAPSHOT.sha256`.

S5 Names in the Released Files

The released files use internal names for the experiments and attacks. The main experiment is the *v3* campaign, and the adapted comparison is the *v4* adaptation. The joint attack is `joint_true_k32`, the activation-only attack is `forward_k32`, the shuffled-gradient and random-candidate controls are `joint_shuffled_k32` and `joint_random_k32`, the backward-only heuristic is `backward`, and SipIt with 32, 64 and 128 steps per token is `sipit_steps32`, `sipit_steps64` and `sipit_steps128`. The BiSR implementation check is the *C-A* route, and the inverter and its refinement are *SIP* and *bf*. The pass rule's current version is `frozen-gate.v2` and the scorecard policy is `scorecard-policy.v3`. The benchmark profiles are `benchmark.split-privacy.v2`, `benchmark.split-integrity-availability.v1`, `benchmark.genai-app.v1` and `benchmark.agentic.v12`, and the observation points of Figure S2 and Table S1 are S1–S14.

S6 Full Module List and Full-Precision Numbers

Attack modules and record. Each run writes one JSON record with the fields `schema`, `config`, `threat_model`, `provenance`, `results` and `summary`. The coverage report marks each threat as COVERED, PARTIAL, UNMEASURED or OUT_OF_SCOPE. The benchmark profiles group the threats: `benchmark.split-privacy.v2` covers the split privacy families with tenant isolation, telemetry leakage and the router channel; `benchmark.split-integrity-availability.v1` covers protocol integrity, malicious client and server, poisoning, resource exhaustion, key abuse and supply chain; `benchmark.genai-app.v1` covers the OWASP LLM list; and `benchmark.agentic.v12` covers the OWASP agentic list and is out of scope by default. Table S14 lists the 64 modules by family; the count is exact at the reviewed revision `submission-candidate-20260922-r2`.

Table S14: Attack families in the reviewed LLM-Attacker taxonomy (64 registered modules at tag `submission-candidate-20260922-r2`). Modes: T = training, I = inference. Entries describe implemented capabilities; Table S15 records each module's evidence tier.

Family	Modules	Modes	Capability / evidence scope
Trained inversion	`latent-probe`, `latent-sensitivity`, `vec2text-inversion`, `seam-inversion`	T, I	Trained and optimisation-based reconstruction; the main experiment and the validated white-box results have their own observations and budgets
Matching attacks	`latent-matching`	T	Centroid and signature matching; token and order results are kept separately
Key / budget solves	`accumulation`, `stale-key`, `sharded`, `leak-accumulation`, `subspace-joint`, `wire-eval`, `seed-disclosure`	T, I	Known-pair budgets, cross-epoch pooling and key-lifetime controls
Known-plaintext	`known-prefix`, `alignment-search`	T, I	Known-prefix solves and realignment under declared label and alignment assumptions
Gradient / output	`gradient-inversion`, `output-inversion`, `label-scoring`, `batch-label-recon`	T, I	Gradient and output reconstruction and gradient-size property scoring
Membership	`membership`, `lira-membership`	T	Likelihood-ratio and relative-likelihood scoring with null-world controls under a local protocol
Structure / blind	`ica-bss`	T, I	Higher-order blind-source separation beyond covariance attacks
Active / malicious	`active-cloud`, `malicious-server`	T, I	Active-server interventions and steering under declared observation
Model-IP theft	`body-extraction`	T, I	Query-driven functional distillation and weight-similarity measurement
Hardware / link	`gpu-memory`, `kv-timing`, `wire-shape`	T, I	Allocator-residue and contention probes under a co-resident-tenant threat model

continued overleaf

Table S14: Attack families in LLM-Attacker (continued).

Family	Modules	Modes	Capability / evidence scope
Reasoning theft	`reasoning-replay`, `trace-inversion`	I	Replay of opaque reasoning blobs and hidden-trace synthesis, scored over an answer-only control
Integrity / availability	`protocol-integrity`, `key-session-abuse`, `resource-exhaustion`, `poisoning-backdoor`, `malicious-client`, `supply-chain`, `replica-integrity`	T, I	Protocol mutation, key-lifecycle abuse, poisoning and availability arms against defended and vulnerable simulators
Tenant isolation / composition	`multi-tenant-isolation`, `artifact-telemetry-leak`, `router-side-channel`, `privacy-composition`, `view-privacy`, `representation-privacy`	T, I	Cross-tenant channels, store canary sweeps, router-trace classification and composition curves
Application profile (GenAI)	`prompt-injection`, `hidden-context-exposure`, `rag-vector-abuse`, `output-sink-injection`, `misinformation-integrity`	I	Injection scored by policy-layer effect; retrieval and grounding checks
Agentic profile (OWASP ASI)	`agent-goal-hijack`, `tool-misuse`, `identity-privilege-abuse`, `agentic-supply-chain`, `unexpected-code-execution`, `memory-context-poisoning`, `insecure-agent-communication`, `cascading-failure`, `human-agent-trust`, `rogue-agent`, `cross-agent-deputy`	I	Seeded agent sandbox with deny-by-default permissions; attacks scored by policy-layer effect, each defence checked by single-flag ablation
Deployment / eval	`remote-fetch`, `mitm-proxy`, `max-effort`, `gradient-hardening`, `two-seam-delegation`, `bisr-baseline`	T, I	Fleet probing, a relay that forwards traffic unchanged and logs a copy, and attacker-optimisation distributions

Validation tiers and artifacts. Table S15 accounts for every module at the reviewed 64-module revision and assigns its primary evidence class. *Reproduced* records agreement with an external reference on fixed inputs; *adapted* records a prior attack family implemented under a declared local contract; *qualification-tested* records a framework-native module run against a real model, capture or deployment; *synthetic* records qualification on a deterministic fixture or simulator; and *unmeasured/helper* identifies orchestration components whose outputs are provenance and budget records. Status `Q-REAL` denotes real-model or deployment qualification, `Q-SYN` synthetic qualification, `PARTIAL` a mixed evidence tier, `UNMEAS` a pending target measurement, and `HELPER` orchestration support. Protected and null outcomes qualify a module when they match their controls. Deployment claims require `Q-REAL` evidence under the corresponding deployment protocol.

Artifact keys keep the matrix readable. `PA/V3` is the shared Paper A/B `v3-campaign/`; `PB/PC`, `PB/DS`, `PB/LH`, `PB/WB`, `PB/BC`, `PB/BA` and `PB/AG` are the current release's positive controls,

domain, horizon, white-box, BiSR conformance, BiSR adaptation and agentic packages listed in Table S12. `P1/AUD` and `P1/E8-ER` refer to the frozen audit and historical qualification artifacts indexed by the Paper 1 release [15]. `T/name` is the named module's deterministic fixture and self-test in the reviewed code tag. The artifact column names evidence, not merely the source file that registers the module.

Table S15: Validation matrix for all 64 reviewed modules. "Expected" is the direction fixed by the module contract; "observed" is the retained qualification outcome, including negative and protected results. Qualification tests whether the implementation behaves as specified; tier and status distinguish real-target, synthetic and helper evidence.

Module	Tier	Model / target	Defence	Expected outcome	Observed outcome	Positive / negative control	Artifact	Status
Reproduced against an external reference (1)								
`bisr-baseline`	Reproduced	GPT-2, one seam	None; public / trained layers	Local port and archived reference emit identical predictions; adapted refinement remains separately budgeted	**3,072/3,072** paired predictions identical; adapted depth-6/public recovery 93.21%	Reference executable on identical inputs; shared joint arm 97.38%	`PB/BC`, `PB/BA`	`Q-REAL`
Adaptations of prior attack families (10)								
`vec2text-inversion`	Adapted	Generated seam world	Gaussian seam noise	Learned decoder beats nearest-neighbour decoding and degrades with noise	94.8% noiseless and 91.1% at σ=2 in the fixture	Nearest-neighbour and majority controls	`T/name`	`Q-SYN`
`seam-inversion`	Adapted	GPT-2 124M	None, noise, mixup, PCA / bottleneck	Undefended recovery exceeds the corpus control; defence-aware path can expose residual leakage	63.25–100% across domains; validated white-box path 18.85–68.59% across depths	Corpus majority; undefended target; matched path checks	`PB/DS`, `PB/WB`, `PB/PC`	`Q-REAL`
`gradient-inversion`	Adapted	Qwen3-0.6B	None	Multi-restart DLG++ measures recovery across shallow and deep boundaries	1.77%, 0.31%, 0.21% at depths 1, 4, 8 over 90 cells	Restarts, depths and random initialisations	`P1/E8-DLG`	`Q-REAL`
`output-inversion`	Adapted	Qwen3-0.6B; Mistral-7B	None / output-side exposure	Output features should permit recovery above a shuffled or majority control when informative	Attack-specific qualification completed on Qwen3-0.6B and Mistral-7B	Shuffled-output and majority controls	`P1/E8-OUT`	`Q-REAL`
`label-scoring`	Adapted	Qwen3-0.6B	Split fine-tuning / FedAvg	Gradient-norm score separates token-presence properties above AUC 0.5	Property AUC 0.979 and 0.984 in the corrected three-seed study	Label shuffle, AUC 0.5 and corrected seed sampler	`P1/E-A4`	`Q-REAL`

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Table S15: Validation matrix for all 64 reviewed modules (continued).

Module	Tier	Model / target	Defence	Expected outcome	Observed outcome	Positive / negative control	Artifact	Status
`batch-label-recon`	Adapted	Generated linear body	None	Planted labels are recovered; null system remains at control	Closed-form fixture exceeds 95%	Planted-label positive and null control	`T/name`	`Q-SYN`
`lira-membership`	Adapted	GPT-2 124M	Undefended positive target	Likelihood-ratio membership score separates IN and OUT records; null world stays at AUC 0.5	Offline positive control AUC **1.00** on 48 IN / 48 OUT records; low-FPR grid is unresolved at this size	Null-world scores and AUC 0.5	`PB/PC`	`Q-REAL`
`body-extraction`	Adapted	Generated teacher / student body	None	Query-driven student improves held-out functional agreement; weight similarity is kept separate	Fixture raises held-out R^2, cosine and logit agreement over an untrained student	Held-out queries and random-student control	`T/name`	`Q-SYN`
`kv-timing`	Adapted	Cache/timing fixture	Cache isolation / cold cache	Warm and cold time-to-first-token distributions separate only when the cache leaks	Warm/cold and speculative-acceptance fixtures separate; hidden-acceptance arm reports INFO	Cold, randomized-prefix and hidden-telemetry controls	`T/name`	`Q-SYN`
`trace-inversion`	Adapted	Generated reasoning tasks	Answer-only interface	Calibration, search and refinement improve trace consistency over an answer-only or surrogate-style control	Refined/best-of-four traces reach consistency 1.0 in the fixture	Answer-only, surrogate style and uncalibrated arms	`T/name`	`Q-SYN`
Framework-native modules qualified on real models, captures or deployments (15)								
`latent-probe`	Qualification	Qwen3-0.6B; 35B-A3B audit tier	Rotated or permuted latents and decoy rows	Calibrated probe detects planted or observable structure and stays at floor after label shuffle	Paper 1 audit exposes the real/decoy partition on every frame; content effect about +0.92 pp	Injected leak, shuffled labels and label-free band	`P1/AUD`, `P1/E8`	`Q-REAL`

continued overleaf

Table S15: Validation matrix for all 64 reviewed modules (continued).

Module	Tier	Model / target	Defence	Expected outcome	Observed outcome	Positive / negative control	Artifact	Status
`latent-sensitivity`	Qualification	Qwen3-0.6B; 35B-A3B audit tier	Chaff, gauge and known-plaintext compromise	Recovery changes with the compromised fraction; zero-compromise arm stays at control	Historical fraction sweeps retained both breaking and non-breaking cells	Zero fraction, random gauge and chaff-retain controls	`P1/E8`	`Q-REAL`
`latent-matching`	Qualification	Qwen3-0.6B; 35B-A3B audit tier	Rotation, decoys and row reordering	Token and order endpoints exceed their own chance controls only when signatures survive	Real battery retained a negative result; synthetic order-and-decoy excess spans 0.52–5.21 points	Chance order, shuffled labels and constant-token control	`P1/AUD`, `T/name`	`Q-REAL`
`accumulation`	Qualification	Qwen/Mistral E-R tiers	Per-epoch rotation; mid-lite	More labelled pairs improve the epoch-local solve; protected contrast remains low	Historical E-R budget sweeps retain both breaking and non-breaking cells; mid-lite remains at 0–0.5%	Label-free band, epoch-local solve and mid-lite	`P1/ER`	`Q-REAL`
`stale-key`	Qualification	Qwen3.5-27B	Per-epoch rotation	Stale keys fail on a new epoch; fresh labelled pairs recover the mapping	Stale recovery 0.94–2.45%; fresh E=1024 recovery 61.20–65.31%	Fresh-key solve and label-free band	`P1/ER4`	`Q-REAL`
`sharded`	Qualification	Qwen3.5-27B; Mistral-NeMo-12B	Static and staggered sharding	A qualified schedule keeps recovery near label-free across blocks and sessions	Static 56.98–67.86%; staggered 47.58–65.13%; independent 12B replication 54.95–65.68%	Label-free band; static versus staggered arms	`P1/ER5`	`Q-REAL`
`wire-eval`	Qualification	Mistral-7B; Mistral-NeMo-12B	Fixed obfuscation and per-epoch ratchet	Protected wire recovery falls while identity-restoration control remains correct	E9 baseline 50% to obfuscated 0%; rotated-epoch wire recovery 0% in the ratchet tier	Unobfuscated baseline, shared-key restoration and RTT twins	`P1/E9`, `P1/ER`	`Q-REAL`
`known-prefix`	Qualification	Qwen3-0.6B; Qwen/Mistral E-R tiers	Fixed or jittered prefix / rotation	More known prefix pairs cross the predeclared recovery fraction only when alignment survives	Qwen depth-1 K_{90} approximately 232–240 for prefixes 32–128; non-crossings retained	Label-free band, prefix-length sweep and jittered arm	`P1/E8-KP`, `P1/ER3`	`Q-REAL`

continued overleaf

Table S15: Validation matrix for all 64 reviewed modules (continued).

Module	Tier	Model / target	Defence	Expected outcome	Observed outcome	Positive / negative control	Artifact	Status
`alignment-search`	Qualification	Mistral-NeMo-12B E-R9 diagnostic	Prompt jitter and bidirectional folding	Realignment lifts recovery when jitter hides correspondence	Approximate alignment reached 15.21% versus 9.71% label-free, but no epoch produced an accepted solve	Unjittered scaffold, label-free band and solved-outcome gate	`P1/ER9`	`PARTIAL`
`membership`	Qualification	Qwen3-0.6B	Split fine-tuning / FedAvg	Corrected seeded sampling yields repeatable membership AUC above 0.5 if features leak	Membership AUC 0.875±0.013 and 0.874±0.008 over three training seeds	Shuffled membership and distinct feature hashes	`P1/E-A4`	`Q-REAL`
`view-privacy`	Qualification	GPT-2 plus exported bundle fixtures	Secret mixup, complementary views, tripwires	Oracle/coalition or repeated views expose more than a single protected view; honest controls remain usable	Bundle and repeated-view qualification completed; separate v3 attacks leave 83.01–90.67% token recovery under mixup	Single view, known-mean oracle, honest correction and reset windows	`PA/V3`, `T/name`	`PARTIAL`
`representation-privacy`	Qualification	GPT-2 plus bottleneck bundles	Trained bottleneck, PCA and Gram flattening	Fresh coordinate/norm/Gram probes detect residual signal; label shuffle remains null	Bundle/probe qualification completed; separate validated white-box attack recovers 18.85–68.59%	Shuffled labels, fresh probes and public/private-phase controls	`PB/WB`, `T/name`	`PARTIAL`
`agent-goal-hijack`	Qualification	Qwen3-0.6B planner + sandbox	Instruction provenance	Vulnerable simulator violates and defended simulator blocks without rejecting sanctioned actions	Synthetic direction reproduced; later live tier measured every setting with no violation but blocked legitimate actions in the sanctioned contrasts	Sanctioned request, no-control and provenance-only arms	`PB/AG`, `T/name`	`PARTIAL`
`cross-agent-deputy`	Qualification	Qwen3-0.6B planner + agent bus	Principal binding versus message authentication	Authentic forwarded request violates without principal binding and is rejected with binding	Synthetic single-flag contrast reproduced; live planner produced no violation and false-blocked sanctioned requests in the measured tier	Sanctioned request and authentication-only distinctness arms	`PB/AG`, `T/name`	`PARTIAL`

continued overleaf

Table S15: Validation matrix for all 64 reviewed modules (continued).

Module	Tier	Model / target	Defence	Expected outcome	Observed outcome	Positive / negative control	Artifact	Status
`gradient-hardening`	Qualification	GPT-2 124M	Gradient noise, clipping and 8/4-bit quantisation	True-gradient recovery remains above shuffled controls until hardening removes useful signal	At depth six/public, σ=0.1 gives 97.31%, 8-bit gives 97.33%, true gradient 97.38%, shuffled 93.91%	True, shuffled and random-candidate arms	`PA/V3`	`Q-REAL`
Synthetic and simulator qualification (35)								
`leak-accumulation`	Synthetic	Generated epoch captures	Per-epoch row leakage / rotation	Persistent row leakage compounds across epochs rather than resetting	Seeded APT fixture produces the expected accumulation curve	Epoch-reset and zero-row-leak controls	`T/name`	`Q-SYN`
`subspace-joint`	Synthetic	Generated shared-basis captures	Per-epoch keys	Joint basis solve needs fewer pairs than independent epoch solves when a basis is shared	Planted shared-basis fixture recovers; independent-basis null removes the advantage	Independent solves and no-shared-basis world	`T/name`	`Q-SYN`
`seed-disclosure`	Synthetic	Generated encoder	Disclosed initialisation seed	Correct seed reproduces the encoder digest and lifts inversion; wrong seed does not	Digest and reconstruction checks pass in the fixture	Wrong/random seed and digest mismatch	`T/name`	`Q-SYN`
`ica-bss`	Synthetic	Generated mixed sources	Unknown rotation / mixing	FastICA separates non-Gaussian sources above chance but not a null mixture	Deterministic ICA fixture qualifies the solver	Permuted-source and null-mixture controls	`T/name`	`Q-SYN`
`active-cloud`	Synthetic	Split-model fixture	Crafted returned tensors / updates	Crafted cloud intervention increases leakage over passive honest behaviour	Planted intervention direction reproduced in the fixture	Honest cloud and zero-perturbation arms	`T/name`	`Q-SYN`
`malicious-server`	Synthetic	Splitbench fixture	Crafted body computation	Crafted amplification makes returned gradients more decodable than honest gradients	Crafted arm exceeds the honest arm at the same fixture budget	Honest server and equal-budget passive arm	`T/name`	`Q-SYN`

continued overleaf

Table S15: Validation matrix for all 64 reviewed modules (continued).

Module	Tier	Model / target	Defence	Expected outcome	Observed outcome	Positive / negative control	Artifact	Status
`gpu-memory`	Synthetic	Allocator / unified-memory fixture	Zero-on-free and isolation	Residue or contention canaries appear in the vulnerable world and disappear after scrubbing/isolation	Canary and contention directions reproduced	Scrubbed buffers and idle/isolation controls	`T/name`	`Q-SYN`
`wire-shape`	Synthetic	Generated packet journal	Padding / shape hiding fixture	Message boundaries, phases and token counts are recoverable only from informative shapes	Clean fixture boundary F1 is 1.0 with low token-count error; null/padded arms bound interpretation	Empty, padded and shuffled-timing journals	`T/name`	`Q-SYN`
`reasoning-replay`	Synthetic	Opaque reasoning-blob fixture	Session/model binding	Compatible replay succeeds only under matching provider/session assumptions	Compatibility and poisoned-blob arms execute	Incompatible model/session and answer-only controls	`T/name`	`Q-SYN`
`protocol-integrity`	Synthetic	Split-protocol simulator	Replay, reorder, downgrade, tamper, oversize	Vulnerable endpoint accepts mutations; defended endpoint rejects them while accepting valid frames	All declared mutation classes reproduce the expected open/closed direction	Untampered valid-frame control	`T/name`	`Q-SYN`
`key-session-abuse`	Synthetic	Key-lifecycle simulator	Nonce reuse, rollback, downgrade, seed retention	Vulnerable lifecycle exposes reuse/blast radius; bound fresh sessions do not	Per-arm abuse outcomes and clean session behaviour retained	Fresh nonce/session, current epoch and no-log controls	`T/name`	`Q-SYN`
`resource-exhaustion`	Synthetic	Serving-stack simulator	Caps, queue fairness and cache limits	Adversarial workloads amplify cost in the vulnerable world; caps contain them without blocking benign load	Nine workload arms reproduce amplification/containment directions	Benign tenant and capped defended world	`T/name`	`Q-SYN`
`poisoning-backdoor`	Synthetic	Softmax/RAG fixtures	Poison filtering and update checks	Attack success rises with poison budget; controls and defences lower it	Eight poison families execute across vulnerable and defended worlds	Zero poison and random corruption	`T/name`	`Q-SYN`

continued overleaf

Table S15: Validation matrix for all 64 reviewed modules (continued).

Module	Tier	Model / target	Defence	Expected outcome	Observed outcome	Positive / negative control	Artifact	Status
`malicious-client`	Synthetic	Shared-body simulator	Isolation and input/update validation	Crafted clients extract or destabilise the vulnerable shared body; isolated world contains effects	Crafted, Sybil and instability arms reproduce their declared directions	Honest client and isolated tenant	`T/name`	`Q-SYN`
`supply-chain`	Synthetic	Artifact registry fixture	Digest, provenance, version and safe-load gates	Permissive registry admits each mutation; enforcing registry rejects it before load	Substitution, tamper, unsigned promotion, rollback, drift and unsafe serialization all reproduce	Pristine signed registry loads with zero false rejection	`T/name`	`Q-SYN`
`replica-integrity`	Synthetic	Replica fixture / in-repo channels	Median, K=2 abort and per-channel keys	One malicious replica is contained; collusion or shared-key compromise exposes the designed boundary	Single-fault, collusion, heterogeneity and disclosure arms execute	All-honest and independently keyed replicas	`T/name`	`Q-SYN`
`multi-tenant-isolation`	Synthetic	Shared-serving simulator	Physical/session isolation	Vulnerable shared state leaks canaries; isolated control does not	Session, batch, cache, KV, error and allocator paths reproduce	Physically isolated negative control	`T/name`	`Q-SYN`
`artifact-telemetry-leak`	Synthetic	Generated stores / logs	Redaction, permissions and retention	Planted secrets are found across encodings; redacted stores avoid false positives	Exact, base64, hex, truncated and token-split canaries and permission checks reproduce	Benign store, redacted canaries and retention deadline	`T/name`	`Q-SYN`
`router-side-channel`	Synthetic	Generated MoE traces	Uniform or random router	Informative routing leaks properties/linkability; null router stays near chance	Property and linkability AUC ≥ 0.9 in the fixture; null near 0.5	Uniform-router and flat-contention controls	`T/name`	`Q-SYN`
`privacy-composition`	Synthetic	Seeded multi-surface artifacts	Independent / interacting leaks	Composition curve preserves marginals and reports synergy only when planted	First-failure budget and joint-versus-strongest metrics reproduce	Duplicate-input, independent and single-surface controls	`T/name`	`Q-SYN`
`prompt-injection`	Synthetic	Application sandbox	Instruction provenance / policy layer	Injection causes an unauthorised effect only in the vulnerable world	Direct, indirect, encoded and cross-session directions reproduce	Benign prompt and defended policy world	`T/name`	`Q-SYN`

continued overleaf

Table S15: Validation matrix for all 64 reviewed modules (continued).

Module	Tier	Model / target	Defence	Expected outcome	Observed outcome	Positive / negative control	Artifact	Status
`hidden-context-exposure`	Synthetic	Application-context fixture	Context isolation / output filtering	Vulnerable world reveals planted context; defended world suppresses it	Exact, semantic, prefix, encoded and multi-turn endpoints execute separately	No-secret and answer-only controls	`T/name`	`Q-SYN`
`rag-vector-abuse`	Synthetic	Vector-store fixture	Tenant filters and provenance	Cross-tenant, poison or retention arms affect vulnerable retrieval but not enforced controls	Retrieval, membership, flooding and citation-substitution arms reproduce	Clean corpus, tenant filter and fresh-retention controls	`T/name`	`Q-SYN`
`output-sink-injection`	Synthetic	Downstream sink fixture	Sink-specific policy layer	Unsafe output causes an effect only without the downstream gate	HTML, SQL, shell, template, URL, path, code and structured sinks reproduce	Benign output and defended sink	`T/name`	`Q-SYN`
`misinformation-integrity`	Synthetic	Authoritative-corpus fixture	Grounding / citation checks	Fabricated or stale claims score worse than grounded answers	Groundedness, contradiction, calibration and decision-harm endpoints execute	Authoritative answer and no-evidence controls	`T/name`	`Q-SYN`
`tool-misuse`	Synthetic	Agent sandbox	Tool scopes and budgets	Dangerous parameters/scope amplification execute only in the vulnerable world; authorised task remains usable	Violation, blocking and zero-false-block directions reproduce	Authorised task and single-control ablations	`T/name`	`Q-SYN`
`identity-privilege-abuse`	Synthetic	Agent sandbox	Principal binding	Principal switch executes without binding and is rejected with binding	Confused-deputy and cross-session arms reproduce	Authorised same-principal request	`T/name`	`Q-SYN`
`agentic-supply-chain`	Synthetic	Agent or component registry	Component provenance	Tampered connector redirects the action only without digest-bound provenance	Bad-digest and trusted-label distinctness cases reproduce	Valid component and provenance-only arm	`T/name`	`Q-SYN`
`unexpected-code-execution`	Synthetic	Canned-execution sandbox	Digest/task-bound execution gate	Unapproved digest executes only in the vulnerable world; approved digest remains usable	Direct, document and approval-swap arms reproduce	Authorised execution and single-flag ablations	`T/name`	`Q-SYN`

continued overleaf

Table S15: Validation matrix for all 64 reviewed modules (continued).

Module	Tier	Model / target	Defence	Expected outcome	Observed outcome	Positive / negative control	Artifact	Status
`memory-context-poisoning`	Synthetic	Persistent-memory sandbox	Memory provenance	Cross-session planted directive changes action only without provenance	Poisoned-memory direction and defended block reproduce	Clean memory and authorised persistence	`T/name`	`Q-SYN`
`insecure-agent-communication`	Synthetic	In-memory agent bus	Message authentication / nonce freshness	Spoof, tamper and replay execute only without endpoint verification	Three communication attacks and seven distinctness worlds reproduce	Valid signed message and fresh nonce	`T/name`	`Q-SYN`
`cascading-failure`	Synthetic	Multi-hop agent pipeline	Per-hop revalidation	Broken propagation chain acts only without revalidation; intact authorised chain remains usable	Broken/intact and owner/attacker distinctness cases reproduce	Intact authorised chain and revalidation-only arm	`T/name`	`Q-SYN`
`human-agent-trust`	Synthetic	Human-approval fixture	Faithful display	Displayed/actual mismatch executes only without display integrity	Mismatch and faithful-display arms reproduce	Faithful summary and authorised refund	`T/name`	`Q-SYN`
`rogue-agent`	Synthetic	Lifecycle sandbox	Lifecycle attestation	Unattested replication, self-modification or persistence executes only without attestation	All three lifecycle classes and authorised spawn distinctness reproduce	Operator-attested lifecycle action	`T/name`	`Q-SYN`
`two-seam-delegation`	Synthetic	Generated two-seam model	Private middle segment	Chained decode beats direct seam-2 decode when the middle is planted; identity middle collapses to the single-seam control	Planted and identity worlds reproduce, including noise arms	Single-seam and identity-middle controls	`T/name`	`Q-SYN`
Orchestration helpers (3)								
`remote-fetch`	Unmeasured / helper	Configured fleet	Reachability and access controls	Reachable captures are copied with provenance; unreachable/disabled nodes are recorded cleanly	Orchestration behaviour and clean unreachable-node handling are tested	Disabled and unreachable node controls	`T/name`	`HELPER`

continued overleaf

Table S15: Validation matrix for all 64 reviewed modules (continued).

Module	Tier	Model / target	Defence	Expected outcome	Observed outcome	Positive / negative control	Artifact	Status
`mitm-proxy`	Unmeasured / helper	Local relay fixture	Transparent forwarding	Forwarded bytes remain unchanged while both directions are journalled	Relay/tee behaviour and byte identity are tested	Direct endpoint and byte-identity check	`T/name`	`HELPER`
`max-effort`	Unmeasured / helper	Wrapped attack	Budget / seed sweep	Wrapper preserves the target attack and reports its distribution over effort	Interface and effort-distribution machinery are qualified	Single-budget wrapped run	`T/name`	`HELPER`

Full-precision numbers. The main experiment's primary difference is 3.1714 percentage points; the Wilson multiplier is $z = 2.5391848136513127$; the bootstrap uses 100,000 replicates with seed 93281. The domain-battery floors round to 3.911, 2.603 and 2.342 percentage points. The main experiment used 336.1 hours of summed online attack time and wrote 17 hash-chained logs. The validated white-box runs each record 128 optimisation steps at learning rate 0.05, a victim learning rate of 5×10^{-5}, 64 training steps, 32 calibration and 32 attacked sequences, 1,024 scored positions, 131,072 token positions in each direction and 51,463,168 vocabulary comparisons in readout, with 1,223.24 seconds of attack time over the 198 stable runs and 617.16 seconds over the 99 co-trained runs. The BiSR implementation check trained six inverters for 20 epochs on 112 auxiliary articles (15,604 windows), ran 24 prediction settings (6,144 document-engine runs) and used an equivalence interval of $(-0.01, 0.01)$ with $t = 2.919985580355516$; every one of the 32 comparisons gave the interval $[0, 0]$, which describes the agreement on these fixed documents. The adapted comparison ran two attack seeds (17, 29) in 80 joint executions, for 160 scored runs and 20,480 document executions, reused 240 control runs, trained four GRU inverters, one per depth and attack seed, for twenty epochs, applied 32 gradient-matching and 800 activation-matching steps, multiplied the stored gradients by 256 to convert the loss normalisation to a per-document mean, and used 29.70 hours of prediction time and 0.58 hours of preparation. Independent audits on two CPU hosts of the 500-step extension verified all 13 execution journals and 12,443 raw files. The live-planner canaries and full run used 234.79 seconds of worker wall time.

Synthetic order-and-decoy battery. Some defences shuffle the order of the activations and add decoy rows. A synthetic battery, with data generated by the framework, ran the matching module against such data for three seeds. In its six matching reports, the highest excess of order recovery over chance ranged from 0.52 to 5.21 points. The matching module is a local supervised baseline, and the battery shows that the framework scores the order of the rows separately from their tokens.

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